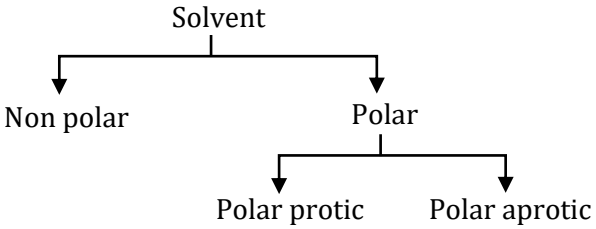
01 Reaction Mechanism

Solvent:

Solvent can be classified as:



Polar solvent:

Polar solvent has some overall dipole moment. They are of two types.

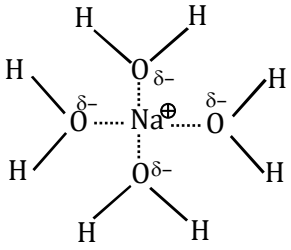
(i) Polar protic solvent : Solvents in which H atom is directly attached with highly electronegative atom. Polar protic solvents are capable of forming intermolecular hydrogen bonding.

In polar protic solvents cation are solvated by ion dipole interaction while an anion is solvated by hydrogen bonding. Polar protic solvent can give $H^{\oplus}$

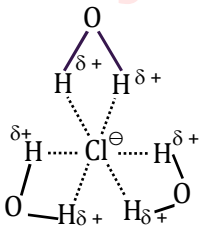
Example : H_2O, C_2H_5OH, CH_3COOH etc.

Polar protic solvent can solvate both cation as well as anion. Cations are solvated by ion dipole interaction with H_2O while anions are solvated by hydrogen bonding with H_2O. Solvation of NaCl in H_2O can be explained as

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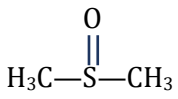
$Na^{\oplus}$ is solvated by ion-dipole interaction with HO_2



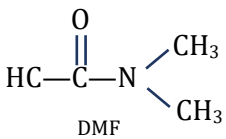
$Cl^{\ominus}$ is solvated by H_2O

(ii) Polar aprotic Solvents : These are the solvents in which H is not attached with highly electro negative atom. These solvents can not give H^+

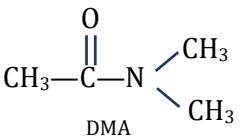
Examples :



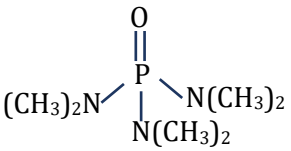
DMSO
(Dimethyl sulphoxide)



DMF
(Dimethyl formamide)



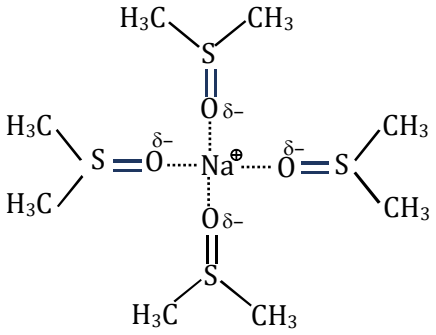
DMA
(Dimethyl acetamide)



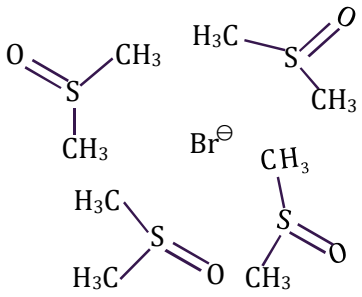
HMPA
(Hexamethylphosphoric triamide)

Polar aprotic solvent can solvate cation very well, while anion is not solvated. When a salt is added then cation is trapped by ion dipole interaction, while anion is not solvated

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$Na^{\oplus}$ is solvated by ion dipole interaction with DMSO



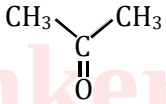
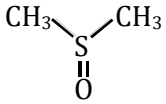
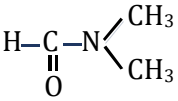
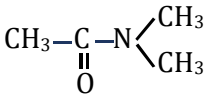
$Br^{\ominus}$ surrounded by solvent but not well solvated by DMSO

Non Polar Solvents:

These solvent has overall zero dipole moment.

Examples : CCl_4, Benzene etc.

Some Common examples of solvents :

S.No.	Solvents	Polar protic	Polar aprotic
1.	H_2O	✓	–
2.	CH_3CH_2OH	✓	–
3.	CH_3CH_2OH	✓	–
4.	CH_3–COOH	✓	–
5.	NH_3	✓	–
6.	 (acetone)	×	✓
7.	 (DMSO)	×	✓
8.	 (DMF)	×	✓
9.	 (DMA)	×	✓

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Illustration 1

Which of the following is a non polar solvent?

(A) H_2O (B)  (C) CCl_4 (D) Ethanol

Ans. (C)

Solution:

H_2O,  & ethanol are polar solvent.

But, in CCl_4, net dipole moment is zero, so non-polar.

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Attacking Reagents:

The species which attack on a substrate molecule or intermediate and form a product is called as attacking reagent.

These are of two types : (1) Electrophiles (2) Nucleophiles

Electrophiles & nucleophiles:

1. Electrophile:

An electrophile is a positively charged species or neutral molecule with electron deficient center. Electrophiles can act as Lewis acids.

Electrophiles are of following type :

(i) Positively charged electrophiles :

$H^{\oplus}$, $Br^{\oplus}$, $Cl^{\oplus}$, $I^{\oplus}$, $NO_2^{\oplus}$, $R_3C^{\oplus}$, $NO^{\oplus}$ etc.

(ii) Neutral Electrophiles : In these electrophiles central atom has an incomplete octet and at least one vacant orbital

SO_3, BF_3, $AlCl_3$, $ZnCl_2$, $FeCl_3$, RCOCl, $(RCO)_2O$

2. Nucleophiles:

It is an electron rich species which has at least one lone pair of electrons. Nucleophile can be neutral or negatively charged. Nucleophile is always a lewis base.

Example: $CN^{\ominus}$, $OH^{\ominus}$, $Br^{\ominus}$, $I^{\ominus}$, NH_3 , H_2O

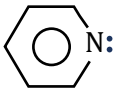
Nucleophile are of following types :

(i) Negatively charged nucleophiles:

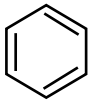
Example: $\overset{\ominus}{H}$, $\overset{\ominus}{O}H$, $\overset{\ominus}{O}—R$, $\overset{\ominus}{C}H_3$, $\overset{\ominus}{X}$, $\overset{\ominus}{S}H$, $R—\overset{\ominus}{S}$, $\overset{\ominus}{N}O_2$, $R—\overset{O}{\overset{\|}{C}}—\overset{\ominus}{O}$, etc

(ii) Neutral nucleophiles: In these nucleophiles central atom has complete octet and at least one lone pair of electron

Example: $R—\ddot{N}H_2$, $\ddot{N}H_3$, $\ddot{N}H_2—\ddot{N}H_2$, $H—\ddot{O}—H$, $R—\ddot{O}—H$, $R—\ddot{O}—R$,

$H—\ddot{S}—H$, $R—\ddot{S}—H$, $R—\ddot{S}—R$, $\ddot{P}H_3$, 

Organic compounds containing carbon-carbon double bonds can also behave as nucleophile.

$CH_2=CH_2$,  , $CH_2=CH—CH=CH_2$, etc.

(iii) Ambident Nucleophile: Species having two nucleophilic sides, but only one can donate lone pair of electron at a time, known as ambident nucleophile.

$C^{\ominus}\equiv\ddot{N}$, $^{\ominus}O—\ddot{N}=O$, etc.

(iv) Amphiphile: Molecule containing a multiple bond between carbon and a more electronegative atom can act both as electrophiles or nucleophiles

$>C=O$ $CH_3—C\equiv N$

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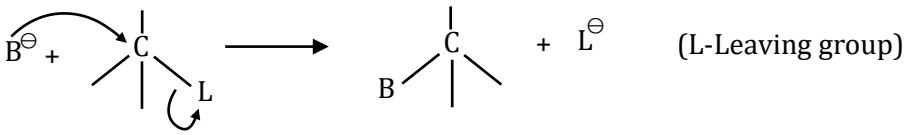
S.No.	Electrophile	Nucleophile
1	Accepts the electron pair	Supplies the electron pair
2	Electron deficient	Electron rich
3	Lewis acid	Lewis base
4	Possess an empty orbital to receive the electron pair	Possess an electron pair which is loosely held and can be supplied easily
5	Usually positively charged species	Usually negatively charged species
6	Forms an extra bond with the nucleophile	Increases its covalency by one unit

Nucleophilicity & basic strength:

Nucleophilicity is defined as the tendency of any species to give electron pair to an electron deficient center, while base strength is the ability of the species to remove $H^{\oplus}$ ion, from an acid

Example : Consider a species $B^{\ominus}$

(i) It functions as a nucleophile is given as



(L-Leaving group)

(ii) It functions as base is indicated as

$$B^{\ominus} + H{-}A \rightleftharpoons B{-}H + A^{\ominus}$$

The nucleophilicity is determined by the kinetics of the reaction, which is reflected by its rate constant (k) while base strength is determined by the equilibrium constant, which is reflected by its K_b.

Criteria for Nucleophilicity: The factors which increases electron density at donor atom increases nucleophilicity.

The more polarisable donor atom is a better nucleophile. Therefore, large size of donor atom increases nucleophilicity.

Effect of the solvent : In polar protic solvent large nucleophiles are good, and the nucleophilicity of halide ions follows the order as :

$$F^{\ominus} < Cl^{\ominus} < Br^{\ominus} < I^{\ominus}$$

In DMSO, the relative order of nucleophilicity of halide ions is given as

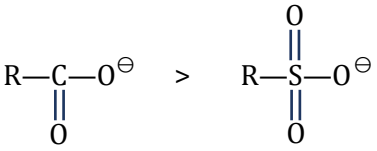
$$F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}$$

Also base strength of halides follows the same order in DMSO but in polar protic solvent like H_2O they follow the reverse order of nucleophilicity.

This effect is related to the strength of the interaction between nucleophile and solvent molecules of polar protic solvent forms hydrogen bond to nucleophiles in the following manner :

Relative nucleophilicity in polar protic solvent

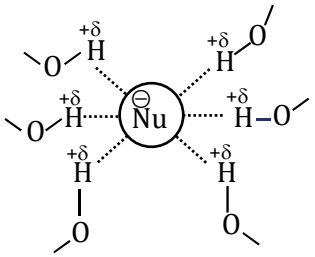
$$\overset{\ominus}{SH} > \overset{\ominus}{I} > \overset{\ominus}{OH} > \overset{\ominus}{N_3} > \overset{\ominus}{Br} > \overset{\ominus}{Cl} > \overset{\ominus}{F} > H_2O$$



$$HO^{\ominus} > H_2O$$

$$NH_2^{\ominus} > NH_3$$

$$CF_3SO_3^{\ominus} < PhCOO^{\ominus} < PhO^{\ominus} < RO^{\ominus}$$

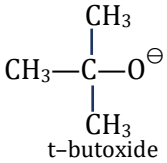


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Steric effects on nucleophilicity:

$CH_3CH_2-O^{\ominus} > (CH_3)_2CH-O^{\ominus} > (CH_3)_3C-O^{\ominus}$ (Nucleophilicity)

Due to steric crowding nucleophilicity decreases.



t–butoxide

(Stronger base, yet weaker nucleophile cannot approach the carbon atom so easily)

$CH_3-CH_2-O^{\ominus}$ (Weaker base, yet stronger nucleophile)

ethoxide

Nucleophilicity : Nucleophilicity decreases from left to right in a period and base strength also decreases.

$CH_3^{\ominus} > NH_2^{\ominus} > OH^{\ominus} > F^{\ominus}$

In a group, nucleophilicity increases from top to bottom due to increases in size of donor atom, but base strength decreases from top to bottom.

Acid strength : HI > HBr > HCl > HF

Base strength : $F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}$

Nucleophilicity : $F^{\ominus} < Cl^{\ominus} < Br^{\ominus} < I^{\ominus}$

It is the tendency to donate electron pair to $H^{\oplus}$ ion.

(a) $NaOH + CH_3-Cl \longrightarrow CH_3OH + Na^{+}Cl^{\ominus}$

(Nucleophile)

(b) $NaOH + HCl \longrightarrow H_2O + Na^{\oplus}Cl^{\ominus}$

(base)

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NUCLEOPHILICITY vs BASIC STRENGTH

S.NO.	Nucleophilicity	Basic strength	Remarks
1.	$CH_3^{\ominus} > NH_2^{\ominus} > OH^{\ominus} > F^{\ominus}$	$CH_3^{\ominus} > NH_2^{\ominus} > OH^{\ominus} > F^{\ominus}$	If donor atoms belong to same period, nucleophilicity and basic strength order are same
2.	$SiH_3^{\ominus} > PH_2^{\ominus} > SH^{\ominus} > Cl^{\ominus}$	$SiH_3^{\ominus} > PH_2^{\ominus} > SH^{\ominus} > Cl^{\ominus}$	If donor atoms belong to same period, nucleophilicity and basic strength order are same
3.	$F^{\ominus} < Cl^{\ominus} < Br^{\ominus} < I^{\ominus}$	$F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}$	In a group nucleophilicity increases while basic strength decreases. on moving top to bottom.
4.	$OH^{\ominus} < SH^{\ominus}$	$OH^{\ominus} > SH^{\ominus}$	In a group nucleophilicity increases while basic strength decreases. on moving top to bottom.
5.	$RO^{\ominus} < RS^{\ominus}$	$RO^{\ominus} > RS^{\ominus}$	In a group nucleophilicity increases while basic strength decreases. on moving top to bottom.

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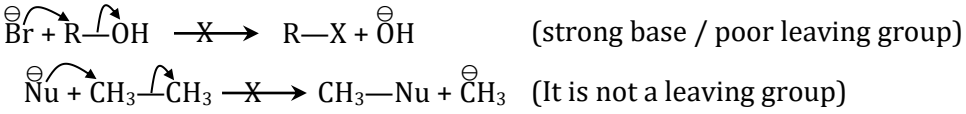
Leaving Group Ability/ Nucleophilicity:

Good leaving groups are those that become the most stable ion after they leave, because leaving group generally leave as a negative ion, so those leaving group are good, which stabilise negative charge most effectively and weak base do this best, so weaker is the base better is the leaving group.

– The leaving group should have lower bond energy with carbon.

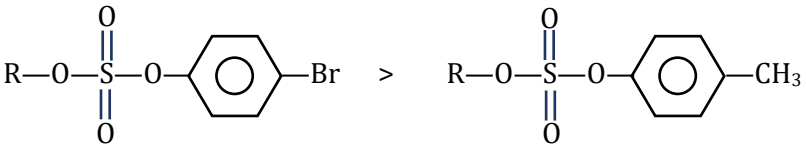
– Negative charge should be more stable either by dispersal or delocalization.

Strongly basic ions rarely act as leaving group →



Order of leaving ability of some groups:

1. $R-\overset{\oplus}{N}\equiv N > R-\overset{\oplus}{O}(H)-R' > R-O-S(=O)(=O)-C_4F_9 > R-O-S(=O)(=O)-CF_3$

2. 

3. $R-O-S(=O)(=O)-C_6H_5 > R-S(=O)(=O)-CH_3 > RI > R-Br > RCl > RF$

4. $R-O-C(=O)-CH_3 > R-O-H > R-O-R > R-H$

where group attached with R is a leaving group

5. $CH_3^{\ominus} < NH_2^{\ominus} < OH^{\ominus} < F^{\ominus}$

6. $R-COO^{\ominus} > PhO^{\ominus} > HO^{\ominus} > RO^{\ominus}$

7. $^{\ominus}SH > {}^{\ominus}OH$

NOTE:

i. Nucleophilicity depends on the nature of solvent.

ii. Anions are better nucleophile than their neutral molecule.

iii. Nucleophilicity increases if adjacent atom of nucleophilic side has lone pair, because lone pairs repels each other. Example $\ddot{N}H_2 - \ddot{N}H_2$.

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Illustration 2

Which of the following species is an electrophilic in nature:-

(A) BH_3 (B) $H_3O^{\oplus}$ (C) $^{\oplus}NO_2$ (D) $Cl^{\oplus}$

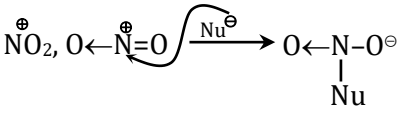
Ans. (A, C, D)

Solution:

(A) BH_3 due to empty p orbital, BH_3 is electrophilic in nature

(B) $H_3O^{\oplus}$ even through contains positive charge but not electrophilic in nature due to unavailable free orbital.

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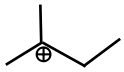
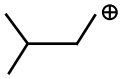
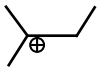
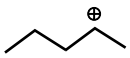
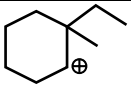
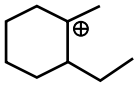
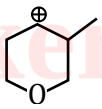
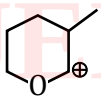
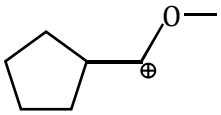
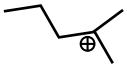
(C) 

So $\overset{\oplus}{N}O_2$ is electrophilic in nature

(D) $Cl^{\oplus}$ due to empty p orbital electrophilic in nature.

Carbocation Rearrangement:

Rearrangement :

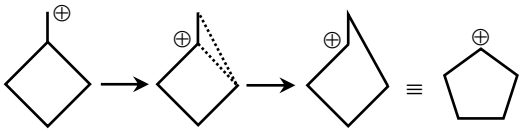
S. No.	Carbocation	No. of 1-2 shift	Most stable
1		1(H)	
2		1(Me)	
3		2(H)	
4		1(H)	
5		1(Et)	
6		2(H)	
7		X	X
8		1(Et)	

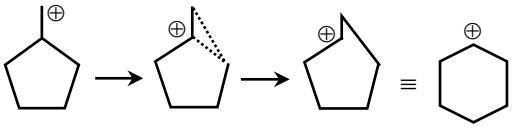
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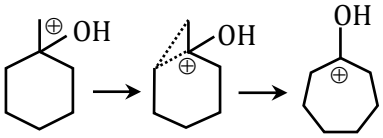
(A) Ring expansion:

It is also an example of 1, 2 shift.

On ring expansion strain get released

(i) 

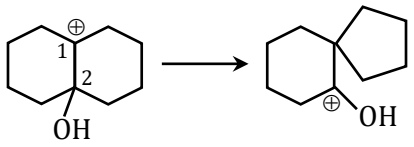
(ii) 

(iii) 

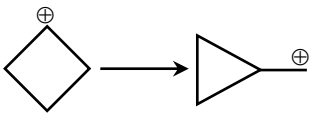
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(B) Ring contraction:

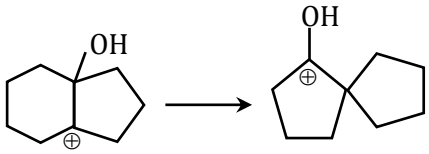
(i)



(ii)



(iii)

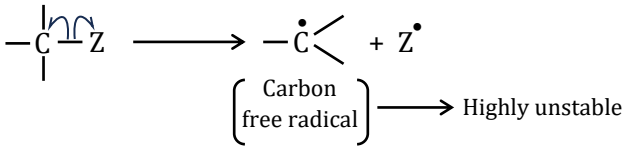


Reaction intermediates:

In the study of organic chemistry following intermediates are more important :

(1) Carbon free radical (2) Carbocation (3) Carbanion
(4) Carbene (5) Nitrene (6) Benzyne

Carbon free Radical:



Characteristics of carbon free radical:

(i) Electrically neutral species.
(ii) $7e^-$ in valence shell
(iii) Octet incomplete
(iv) Electron deficient species
(v) Act as an electrophile
(vi) Neither a Lewis acid nor a lewis base
(vii) 3σ-bond, zero lone pair, zero vacant orbital
(viii) 1 unpaired electron, paramagnetic in nature
(ix) Hybridisation sp^2, trigonal planar
(x) No rearrangement
(xi) Stability of carbon free radical depends on resonance effect, hyperconjugation, inductive effect.

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Stability of carbon free radical:

Ex. $H_3C-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C^{\bullet}}} > H_3C-\dot{C}H-CH_3 > H_3C-\dot{C}H_2 > \dot{C}H_3$ (stability order)

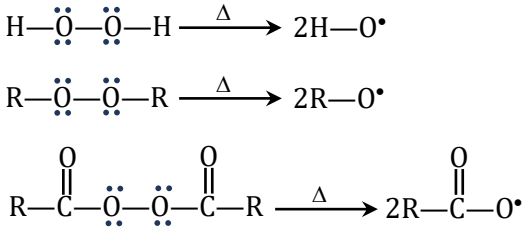
Ex. $(C_6H_5)_3C^{\bullet} > (C_6H_5)_2\dot{C}H > C_6H_5-\dot{C}H_2 > CH_2=CH-\underset{\underset{CH_3}{|}}{\dot{C}}-CH_3$ (stability order)

Some general method of generation of the carbon free radical:

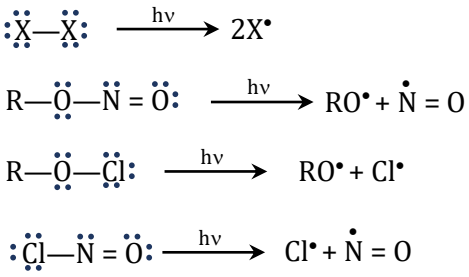
(1) By using heat
(2) By using sunlight
(3) By using metal or metal salt

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(1) By using heat



(2) By using sunlight



(3) By using metal or metal salt

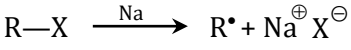


Illustration 3

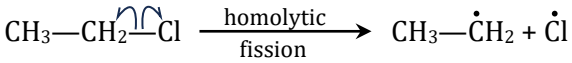
$CH_3CH_2CH_2Cl$ undergoes homolytic fission to produce:-

(A) $CH_3\dot{C}H_2CH_2$ and $\dot{C}l$ (B) $CH_3CH_2\dot{C}H_2$ and $\overset{\oplus}{C}l$

(C) $CH_3CH_2\overset{\oplus}{C}H_2$ and $\dot{C}l$ (D) None of these

Ans. (A)

Solution:



General reactions of carbon free radical:

(a) Combination with a free radical

(i) $\dot{C}H_3 + \dot{C}H_2 - CH_2 - CH_3 \longrightarrow CH_3 - CH_2 - CH_2 - CH_3$

(ii) $CH_3 - \dot{C}H_2 + \dot{C}H_2 - CH_3 \longrightarrow CH_3 - CH_2 - CH_2 - CH_3$

(b) Disproportionate reactions

(i) $CH_2(H) - \dot{C}H_2 + CH_3 - \dot{C}H_2 \longrightarrow CH_2 = CH_2 + CH_3 - CH_3$

(ii) $2CH_3 - \dot{C}H - CH_3 \longrightarrow CH_3 - CH = CH_2 + CH_3 - CH_2 - CH_3$

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Some important reactions of carbon free radical :

Reaction : 1 (Wurtz reaction)

A solution of alkyl halide in ether on heating with sodium gives alkane.

$$R-X + 2Na + X-R \xrightarrow[\text{ether}]{\text{Dry}} R-R + 2NaX$$

(a) Methane can not be prepared by this method. The alkane produced is higher and symmetrical i.e. it contains double the number of carbon atoms present in the alkyl halide taken.

(b) Two different alkyl halides, through Wurtz reaction give all possible alkanes.

(c) The separation of mixture into individual members is not easy because their B.P. are near to each other and thus Wurtz reaction is not suitable for the synthesis of alkanes containing odd number of carbon atom.

(d) This reaction generally fails with tertiary alkyl halide.

Mechanism : Two mechanism have been proposed for this reaction.

(a) Ionic Mechanism:

$$2Na \longrightarrow 2Na^{\oplus} + 2e^{\ominus}$$

$$\overset{\delta+}{R}-\overset{\delta-}{X} + R^{\ominus} \longrightarrow \underset{\text{Product}}{R-R} + X^{\ominus}$$

$$\overset{\delta+}{R}-\overset{\delta-}{X} \xrightarrow{2e^{\ominus}} R^{\ominus} + X^{\ominus}$$

$$2Na^{\oplus} + 2X^{\ominus} \longrightarrow 2NaX$$

(b) Free radical mechanism:

$$Na \longrightarrow Na^{\oplus} + e^{\ominus}$$

$$\dot{R} + \dot{R} \longrightarrow \underset{\text{Product}}{R-R}$$

$$\overset{\delta+}{R}-\overset{\delta-}{X} \xrightarrow{e^{\ominus}} \dot{R} + X^{\ominus}$$

$$Na^{\oplus} + X^{\ominus} \longrightarrow NaX$$

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Illustration 4

Which of the following compound cannot be obtained from single alkyl halide by Wurtz reaction.

(A) Ethane (B) Butane (C) Isobutane (D) Hexane

Ans. (C)

Solution:

Wurtz reaction is suitable for symmetrical alkane preparation containing even number of carbon atoms.

Reaction : 2 (Photohalogenation of alkane)

Halogenation : Replacement of H-atom by halogen atom

$$R-H + X_2 \longrightarrow R-X + HX$$

Halogenation is made on exposure to (halogen + alkane) mixture to UV or at elevated temp.

The reactivity order for halogens shows the order. $\boxed{F_2 > Cl_2 > Br_2 > I_2}$

Reactivity order of hydrogen atom in alkane is $\boxed{\text{Tertiary C}-\text{H} > \text{Sec. C}-\text{H} > \text{primary C}-\text{H}}$

(i) Fluorination : Reacts explosively even in dark. Fluorination can be achieved without violence when alkane is treated with F_2 diluted with an inert gas (like N_2)

By the action of HgF_2 on bromo or iodo derivatives.

$$C_2H_5I + HgF_2 \longrightarrow C_2H_5F + HgI_2$$

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(ii) Chlorination :

The monochloro derivative of alkane is obtained by taking alkane in large excess.

When chlorine is in excess, a mixture of mono, di, tri, tetra and perchloro derivatives is obtained.

$$CH_4 \xrightarrow{Cl_2} CH_3Cl \xrightarrow{Cl_2} CH_2Cl_2 \xrightarrow{Cl_2} CHCl_3 \xrightarrow{Cl_2} CCl_4$$

$$\boxed{CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl}$$

Mechanism for $CH_4 + Cl_2 \xrightarrow{UV} CH_3Cl + HCl$

Step-I Chain initiation step : $Cl - Cl \xrightarrow[or\,\Delta]{UV} \dot{Cl} + \dot{Cl}$

Step-II Chain propagation step : $\dot{Cl} + H\!-\!CH_3 \longrightarrow H\!-\!Cl + \dot{C}H_3$ (Methane; Methyl radical)

$$\dot{C}H_3 + Cl\!-\!Cl \longrightarrow CH_3Cl + \dot{Cl}$$

Step-III Chain termination step : $\dot{Cl} + \dot{Cl} \longrightarrow Cl_2$, $\dot{C}H_3 + \dot{Cl} \longrightarrow CH_3Cl$,

$$\dot{C}H_3 + \dot{C}H_3 \longrightarrow CH_3CH_3$$

(iii) Bromination : Br_2 reacts with alkanes in a similar manner but less vigorously.

(iv) Iodination : Iodine reacts with alkanes reversibly. HI formed as the by product, is a powerful reducing agent and is capable of reducing the CH_3I to CH_4.

Iodination may be carried out in the presence of an oxidising agent such as HIO_3, HIO_4, HNO_3, HgO etc. Which destroy HI,

$$CH_4 + I_2 \rightleftharpoons CH_3I + HI$$

$$5HI + HIO_3 \longrightarrow 3I_2 + 3H_2O$$

Iodination is very slow because energy of activation of the reaction is very large

$$CH_4 + \dot{I} \longrightarrow HI + \dot{C}H_3$$

Reactivity selectivity Principle:

(i) Probability factor: The factor is based on the number of each kind of H atom in the molecule. For example, is $CH_3\!-\!CH_2\!-\!CH_2\!-\!CH_3$ there are six equivalent 1° H's and four equivalent 2° H's. The probability of abstracting 1° H's to 2° H's is 6 to 4. i.e., 3 to 2.

(ii) Reactivity of halogen free radical: the more reactive chlorine free radical is less selective and more influenced by the probability factor. On the other hand, the less reactive Br radical is more selective and less influenced by the probability factor (Reactivity selectivity principle).

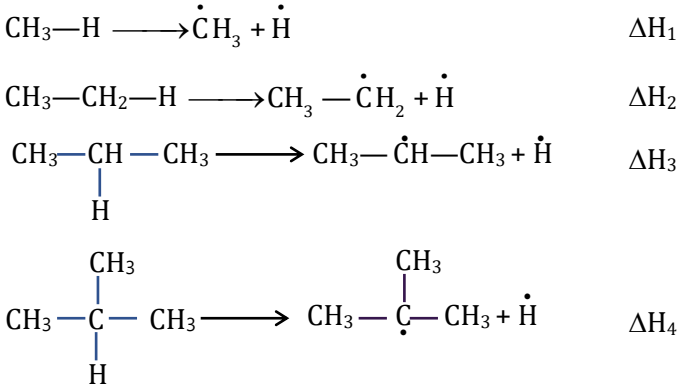
(iii) Reactivity of alkanes (ease of abstraction of 'H' atoms) : Since the rate determining step in halogenations is abstraction of hydrogen by a halogen atom be the formation of alkyl radical, halogenation of alkanes follows order of stability of free radical is 3° > 2° > 1° > CH_3.

Reactivity ratio of H atom for **Chlorination** (1° : 2° : 3° H)
1 : 3.8 : 5

Reactivity ratio of H atom for **bromination** (1 : 82 : 1600)

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The above order of stability of radicals is due to the ease of their formation from the corresponding alkane which in turn is due to difference in the value of ΔH.



Reactivity of any H-atom ∝ number of H atoms of that kind × reactivity of that H.

Thus, the amount of energy required to form the various classes of radicals decreases in the order $CH_3 > 1° > 2° > 3°$ ($\Delta H_1 > \Delta H_2 > \Delta H_3 > \Delta H_4$). Therefore, it is easiest to form 3° radical and it is most difficult to form CH_3. We can also interpret this in an alternative way the case of abstraction of H atoms from hydrocarbon follows the sequence $3° > 2° > 1° > CH_4$ which should also be the case of formation of free radicals.

The above order of stability is in accordance with the stability of free radicals on the basis of delocalization of odd electron. Order of stability of free radical is :

Allyl > benzyl > 3° > 2° > 1° > methyl > vinyl.

Illustration 5

The reaction conditions leading to best yield of C_2H_5Cl are :-

(A) $\underset{\text{(excess)}}{C_2H_6} + Cl_2 \xrightarrow{\text{UV light}}$ (B) $CH_3 + Cl_2 \xrightarrow{\text{UV light}}$

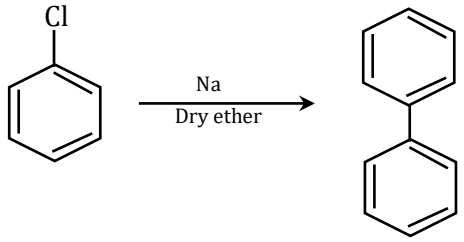
(C) $C_2H_6 + Cl_2\text{(excess)} \xrightarrow{\text{UV light}}$ (D) All of these

Ans. (A)

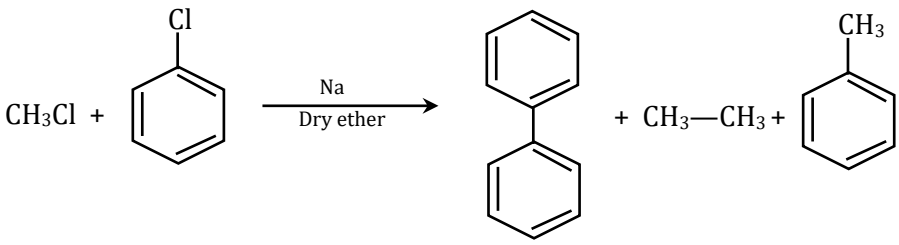
Solution:

$\underset{\text{(excess)}}{CH_3 - CH_3} + Cl_2 \xrightarrow{UV} CH_3 - CH_2 - Cl$

Reaction : 3 (Fittig reaction)

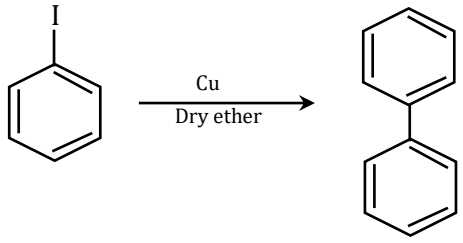


Reaction : 4 (Wurtz-fittig reaction)

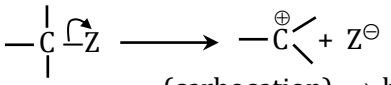


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Reaction : 5 (Ulmann reaction)



Carbocation:



(carbocation) → highly unstable

Characteristics of carbocation

(i) Positively charged species

(ii) $6e^-$ in valence shell

(iii) Octet incomplete

(iv) Electron deficient species

(v) Act as an electrophile

(vi) Lewis acid (due to vacant orbital)

(vii) 3σ -bond, zero lone pair, one vacant orbital

(viii) Zero unpaired electron, diamagnetic in nature

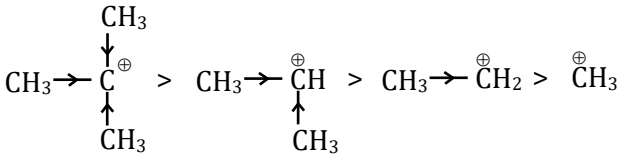
(ix) Hybridisation sp^2, trigonal planar

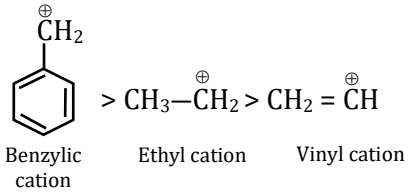
(x) Rearrangement can be possible if stability increases

(xi) Stability of carbocation depends on resonance effect, hyper conjugation effect, inductive effect.

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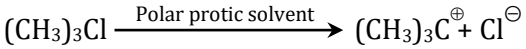
Stability of carbocation

Ex. 

Ex. 

Some general methods of generation of carbocation:

(a) By hetrolytic bond fission of sigma bond.



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(b) By diazotisation

$$CH_3CH_2-NH_2 \xrightarrow{NaNO_2 + \text{ dil. } HCl} CH_3\overset{\oplus}{C}H_2 + N_2$$

(c) By hetrolytic bond fission of π bond.

$$CH_2 = CH_2 + D^{\oplus} \longrightarrow \overset{\oplus}{C}H_2 - CH_2D$$

$$CH_2 = CH - CH_3 + D^{\oplus} \longrightarrow CH_2D - \overset{\oplus}{C}H - CH_3$$

Some general reactions of carbocation:

(a) Combination with a nucleophiles

(i) $CH_3-\overset{\oplus}{C}H_2 \xrightarrow{Cl^{\ominus}} CH_3-CH_2Cl$

(ii) $CH_3-\overset{\oplus}{C}H-CH_3 \xrightarrow{Cl^{\ominus}} CH_3-\underset{Cl}{CH}-CH_3$

(b) Rearrangement:

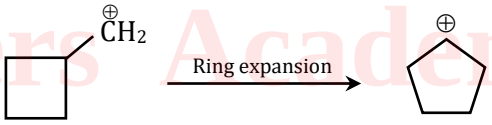
(i) Hydride Shift: $CH_3-\overset{H}{\overset{|}{C}}H-\overset{\oplus}{C}H_2 \longrightarrow CH_3-\overset{\oplus}{C}H-CH_3$

(ii) Methyl Shift: $CH_3-\overset{CH_3}{\overset{|}{C}}-\overset{\oplus}{C}H_2 \longrightarrow CH_3-\underset{\oplus}{\overset{CH_3}{\overset{|}{C}}}-CH_2-CH_3$

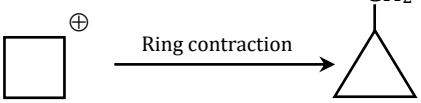
(iii) Aryl Shift: $(CH_3)_2\overset{Ph}{\overset{|}{C}}-\overset{\oplus}{C}H_2 \longrightarrow CH_3-\underset{\oplus}{\overset{CH_3}{\overset{|}{C}}}-CH_2-C_6H_5$

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(iv) Ring expansion:



(v) Ring contraction:



(c) Elimination

$$H-\underset{H}{\overset{H}{\overset{|}{C}}}-\overset{\oplus}{C}H_2 \xrightarrow{-H^{\oplus}} H-C=CH_2$$

Illustration 6

Most stable carbocation is

(A) $CH_3-\overset{\oplus}{C}H_2$ (B) $\overset{\oplus}{C}H_2CHCl_2$ (C) $Cl-CH_2-\overset{\oplus}{C}H_2$ (D) $F-CH_2-\overset{\oplus}{C}H_2$

Ans. (A)

Solution:

(A) $CH_3-\overset{\oplus}{C}H_2$ ($+I$)

(B) $H_2\overset{\oplus}{C}-\underset{Cl}{CH}$ ($-I$)

(C) $Cl-CH_2-\overset{\oplus}{C}H_2$ ($-I$)

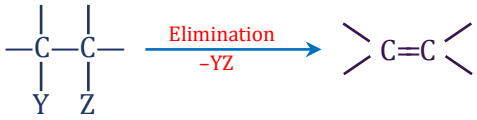
(D) $F-CH_2-\overset{\oplus}{C}H_2$ ($-I$)

Stability of $C^{\oplus} \propto \frac{(+I)}{(-I)}$

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Elimination Reactions:

In elimination reaction two atoms or groups (YZ) are removed from the substrate and generally resulting into formation of π bond.



Elimination can be of following types :

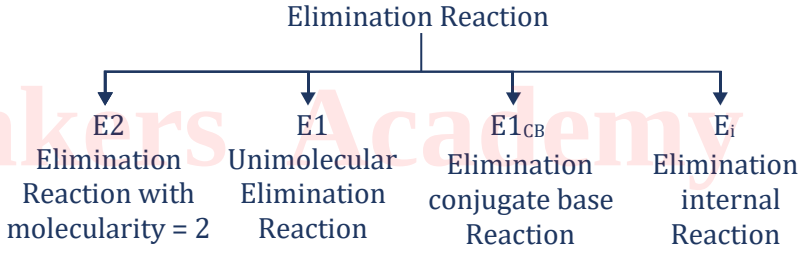
(i) α-elimination: When two groups or atoms are lost from the same carbon it is known as α elimination. It gives a carbene (or nitrene). This is also called 1–1 elimination.



(ii) β-Elimination : When two groups or atoms are lost from adjacent atoms so that a new π bond is formed. This is also called 1–2 elimination.



Based on mechanism, elimination Reactions can be classified as following :



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Bimolecular elimination reaction (E2):

Dehydrohalogenation is the elimination of a hydrogen and a halogen from an alkyl halide to form an alkene.

Example : $CH_3CH_2Cl \xrightarrow{\text{Alc. KOH}} CH_2{=}CH_2$

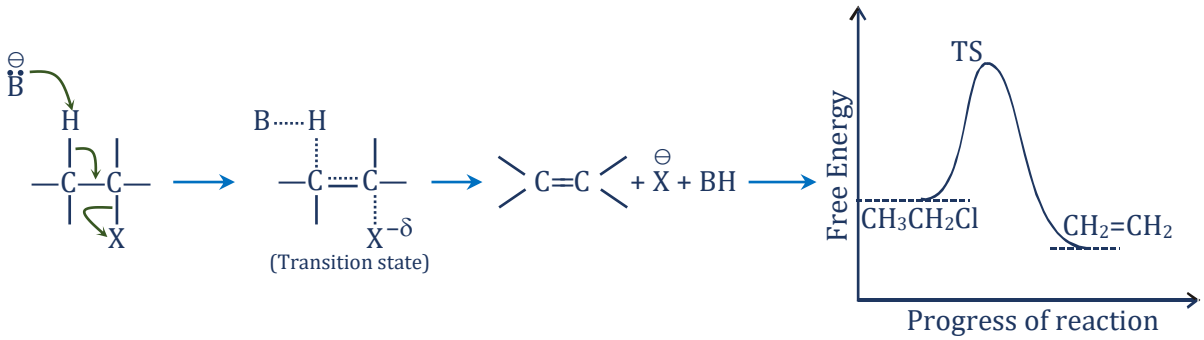
Reagent Used :

(i) alcoholic solution of KOH or $EtO^{\ominus}$/EtOH

(ii) $NaNH_2$

(iii)t-$BuO^{\ominus}$ $K^{\oplus}$ in t-BuOH

Mechanism :



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(i) Characteristics of E_2 reaction :

(a) This is a single step, bimolecular reaction and follows a concerted mechanism.

(b) It is a second order reaction and the kinetics of the reaction can be given as

$$\text{Rate} \propto [R - X]\,[\text{Base}]\ ;\ \text{Rate} = k\,[R - X]\,[\overset{\ominus}{B}]$$

(c) Rearrangement is not possible, since carbocation is not formed.

(d) The orientation of proton & leaving group should be antiperiplanar, i.e., they should be anti to each other or at the angle of 180 to each other.

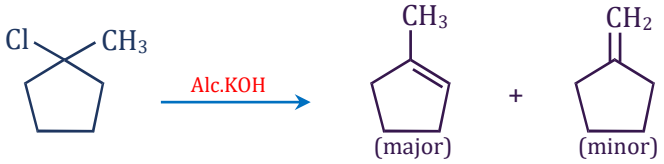
(e) Reactivity order for alkyl halide towards E_2 reaction is given as

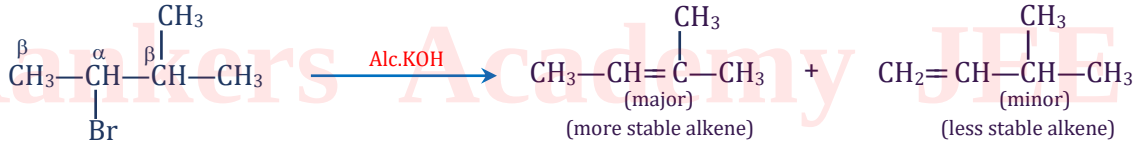
$R - I > R - Br > R - Cl > R - F$

(f) Positional orientation of elimination $\rightarrow$ In most E_2 eliminations where there are two or more possible elimination products.

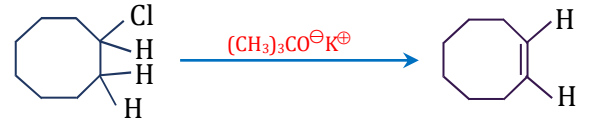
(ii) Formation of saytzeff product: The product with the most highly substituted double bond will predominate. This rule is called the saytzeff or zaitsev rule.

Example :

(a) 

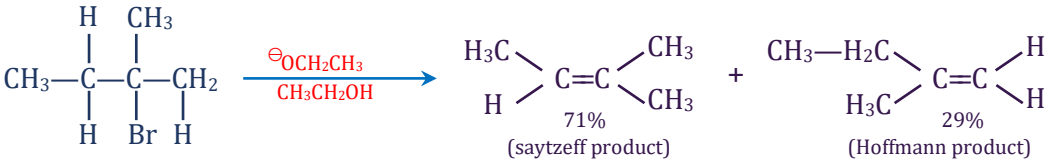
(b) 

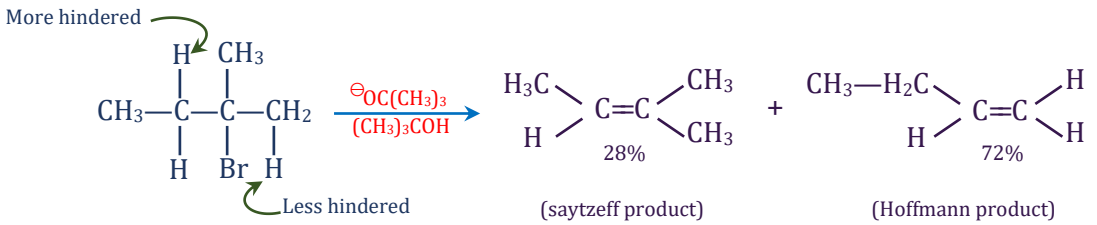
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(c) 

Chlorocyclooctane Cyclooctene

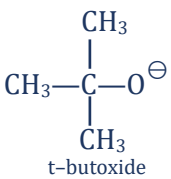
(iii) Formation of the Hoffmann product : Bulky bases can also accomplish dehydrohalogenations that do not follow the saytzeff rule. Due to steric hindrance, a bulky base abstracts a less hindered proton, often that leads to formation of least substituted product, called the Hoffmann product

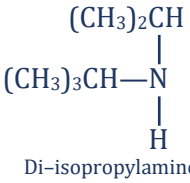


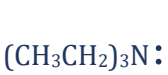


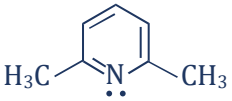
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Bulky base:

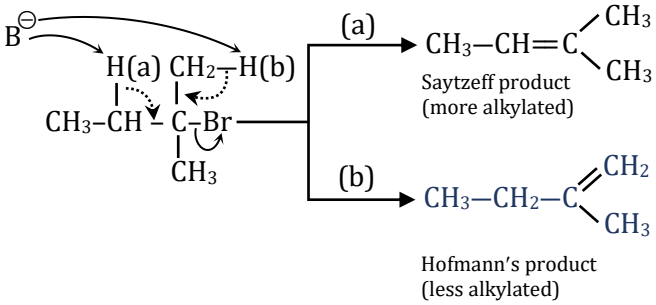

t–butoxide

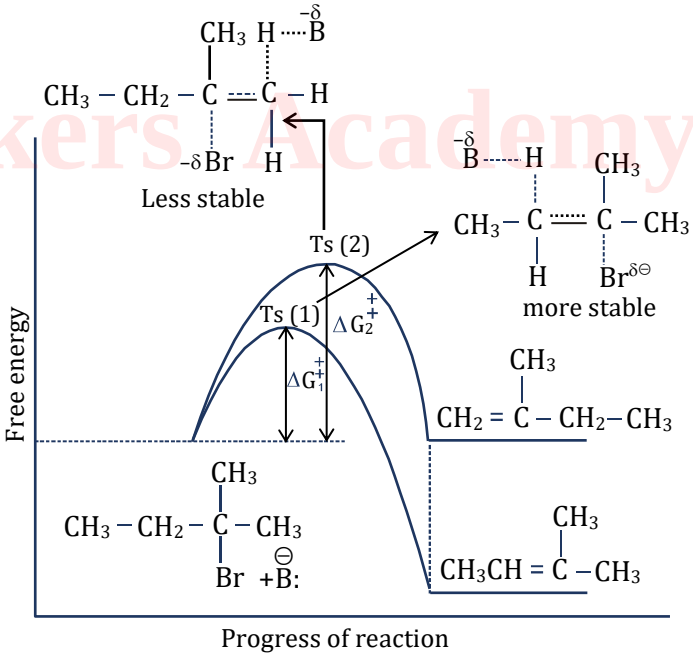

Di–isopropylamine


Triethylamine


2,6–dimethyl pyridine

Example : Dehydrohalogenation of 2–bromo–2–methylbutane can yield two products.





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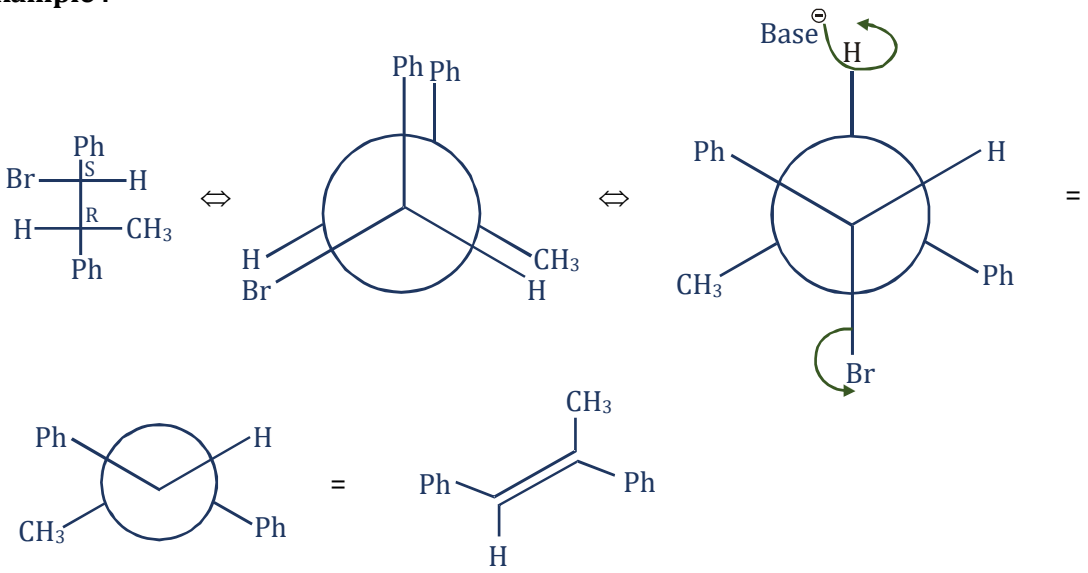
(iv) Stereo Chemistry of E_2 reactions :

E_2 reaction is an example of anti-elimination in which both H and leaving group are antiperiplanar to each other.

The E_2 is stereospecific because it normally goes through an anti periplanar transition state. The stereo specific products are alkene. Formation of product in E_2 can be explained as following:

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Example :



(v) Factors affecting E_2 reactions:

(a) Structure of substrate: Branching at α and β carbon increase rate of E_2 reaction. As number of substituents increase stability of transition state increases. Hence order of reactivity of alkyl halides towards E_2 reaction is 3° > 2° > 1°

(b) Nature of leaving group : Better is the leaving group higher is the rate of E_2 reaction

(c) Strength of the base : Since RDS involves base hence increase in base strength of increase rate of E_2.

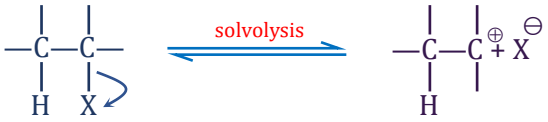
(d) Nature of solvent: The yield of the product formation in E_2 reaction increases with decrease in solvent polarity.

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Unimolecular elimination reaction (E_1):

Proton and leaving group depart in two different step.
First step : Slow step involves ionisation to form carbocation.
Second step : Abstraction of proton.
Mechanism :

Step 1 :



Step 2 :

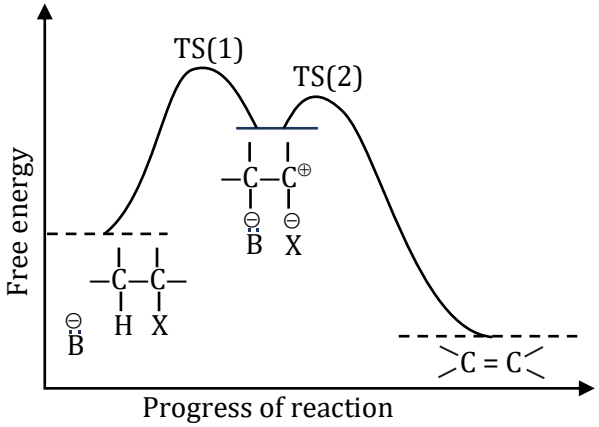


(i) Characteristics of E_1 reaction:

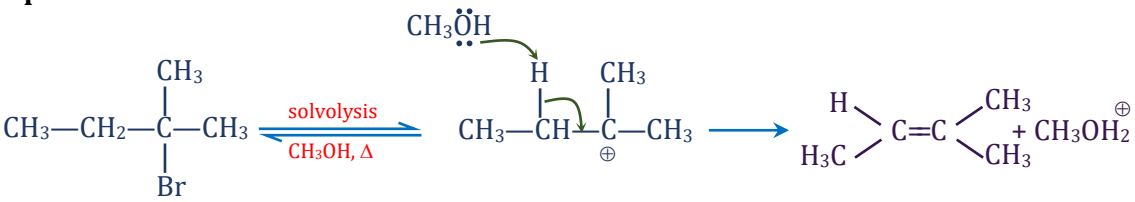
(a) It is unimolecular, two step process.
(b) It is a first order reaction.
Rate ∝ [Alkyl halide]
Rate = k [Alkyl halide]
(c) Reaction intermediate is carbocation, so rearrangement is possible
(d) In the second step, a base abstracts a proton from the carbon atom adjacent to the carbocation, and forms alkene.

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(ii) Energetics of E_1 reaction:



Example :

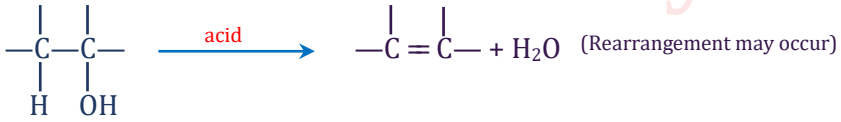


(iii) E_1 Reaction of Alcohol : Dehydration requires an acidic catalyst to protonate the hydroxyl group of the alcohol and convert it to a good leaving group. Loss of water, followed by loss of proton, gives the alkene. An equilibrium is established between reactants and products. For E_1 mechanism reagents are

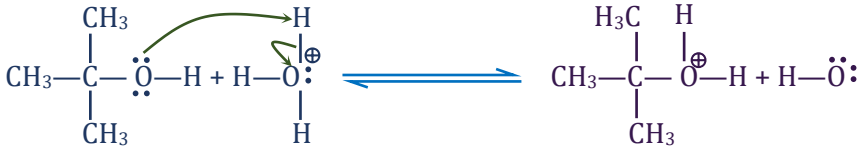
(1) H_3PO_4/Δ (2) H_2SO_4 / 160°

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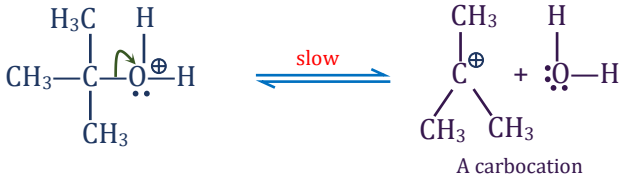
Mechanism :



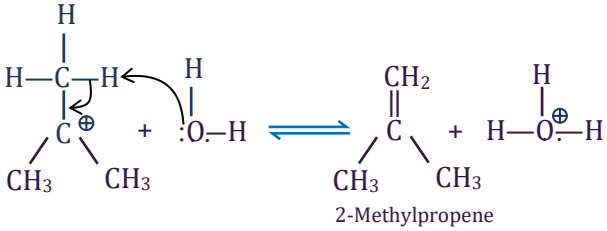
Step 1 :



Step 2 :



Step 3 :



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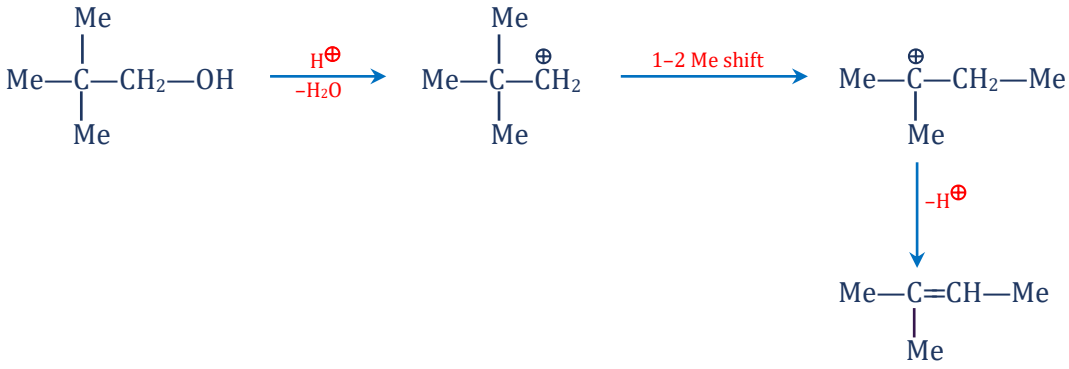
In first step, an acid-base reaction occurs, as a proton rapidly transferred from the acid to one of the unshared electron pairs of oxygen atom of the alcohol.

In second step the carbon oxygen bond breaks. The leaving group is molecule of water :

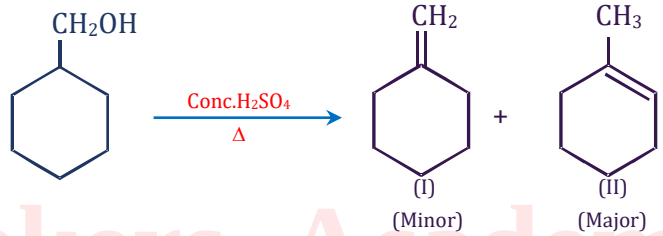
Finally, in third step the carbocation transfers a proton to a molecule of water. The results in the formation of a hydronium ion and an alkene.

Reactivity order of ROH : 3° alcohol > 2° alcohol > 1° alcohol

Example:

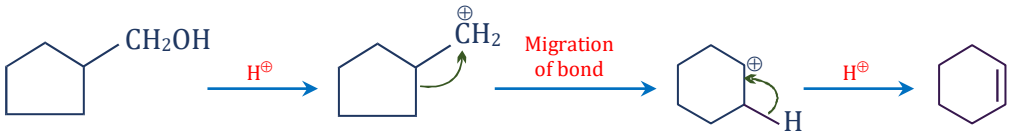


Example:



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Example:



(iv) Factors Affecting E_1 Elimination Reactions :

(a) Structure of Substrate :

Since rate determining step involves formation of carbocation hence order of reactivity will be 3° > 2° > 1° alkyl halide. Alkyl and aryl substitutions on α and β -carbons with respect to the leaving group increase the rate of E_1 reactions. As the strain increases the yield of the E_1 product increases.

(b) Strength and concentration of the base :

Since E_1 reactions do not usually required any base in RDS on increasing the strength and concentration of the base have nothing to do with rate of reaction

(c) Nature of the leaving group :

Better is the leaving group more easier will be formation of carbocation.

(d) Nature of solvent : Greater is the polarity of solvent, more will be rate of E_1

(e) Temperature : With rise in temperature E_1/S_N1 ratio rises, as in the elimination process.

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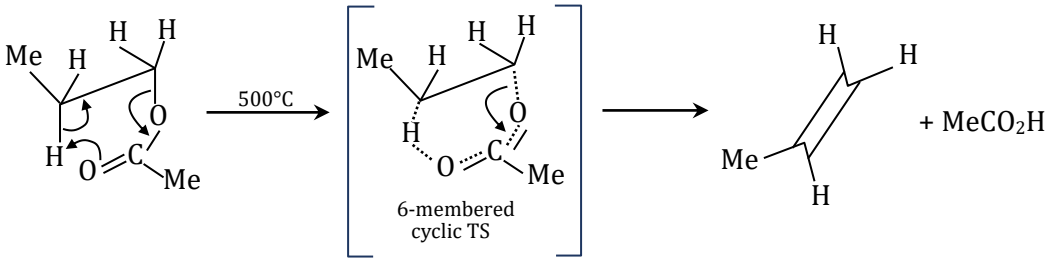
Elimination Internal (Ei) or Pyrolytic syn-Elimination:

These elimination reaction occur through formation of cyclic transition state involving only one molecule of substrate. These elimination reactions occur within a small family of compounds.

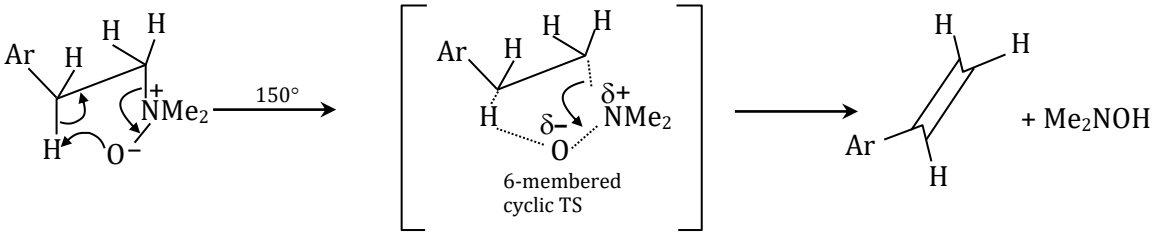
Like acetates, amine oxides sulphoxides, xanthates etc.

Example:

(i) Pyrolysis of esters:



(ii) Cope's reaction:



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Both of the above reactions take place through formation of 5/6 membered cyclic TS and rate is found to be Rate = k[Substrate]. The reactions are possible because the leaving groups are lying in the same plane and on the same side. The elimination process have synthetic utility in producing terminal unrearranged alkene. The above elimination is thus steroselective.

Illustration 7

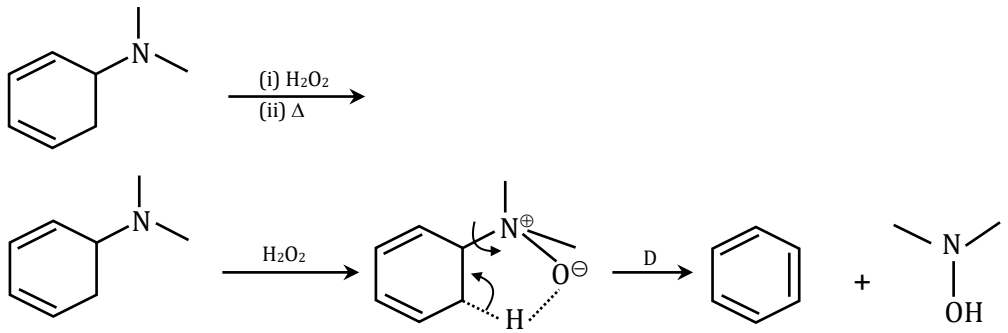
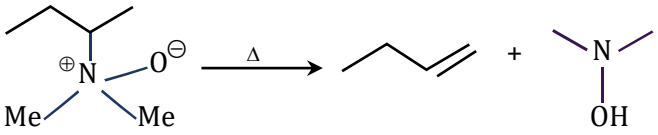


Illustration 8



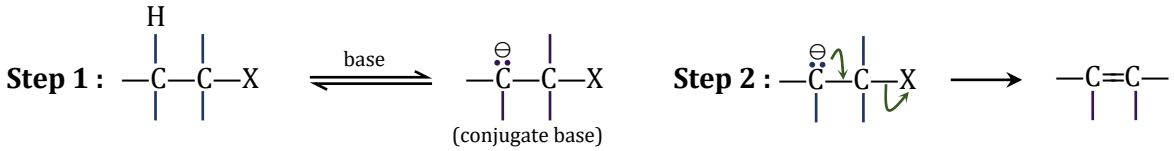
Hoffmann elimination

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Unimolecular conjugate base reaction (E_1cb reaction):

In the E_1CB, H leaves first and then the X. This is a two step process, the intermediate is a carbanion.

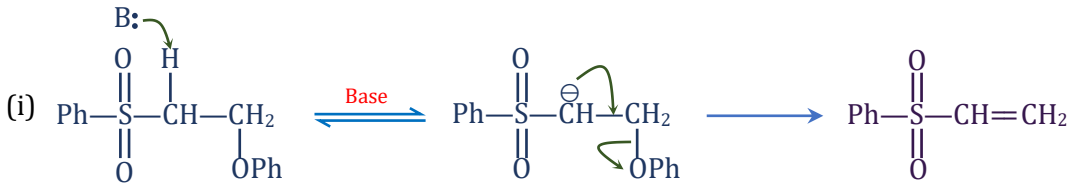
Mechanism :

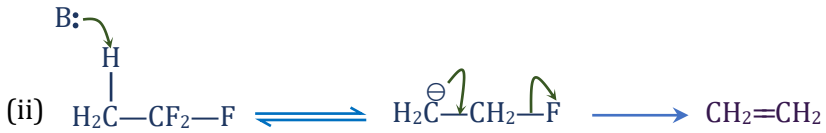


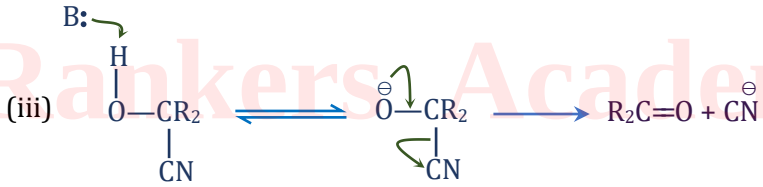
First step consists of the removal of a proton, $H^{\oplus}$, by a base generating a carbanion

In second step carbanion looses a leaving group to form alkene

Condition : For E_1 CB, (i) substrate must be containing acidic hydrogens and (ii) poor leaving groups.







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Illustration 9

In the reaction $CH_3CH_2-\underset{Br}{\overset{CH_3}{\overset{|}{C}}}-\overset{CH_3}{\overset{|}{C}}H-CH_3 \xrightarrow[C_2H_5OH]{C_2H_5ONa}$ the major product obtained is :

(A) $CH_3CH=\overset{CH_3}{\overset{|}{C}}-CH(CH_3)_2$

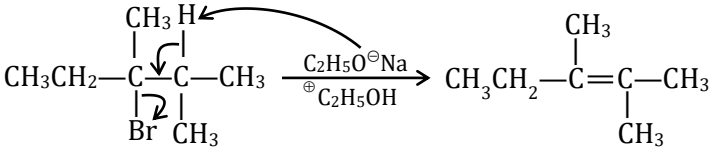
(B) $CH_3CH_2-\underset{OC_2H_5}{\overset{CH_3}{\overset{|}{C}}}-CH(CH_3)_2$

(C) $CH_3CH_2-\overset{CH_3}{\overset{|}{C}}=\overset{CH_3}{\overset{|}{C}}-CH_3$

(D) $CH_3CH_2-\overset{CH_2}{\overset{||}{C}}-CH(CH_3)_2$

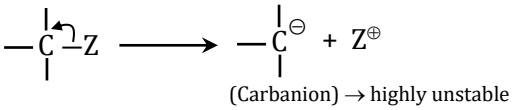
Ans. (C)

Solution:



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Carbanion:

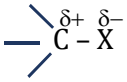


(Carbanion) → highly unstable

Nucleophilic substitution reaction (s_n):

Nature of c–x bond :

Since halogen atoms are more electronegative than carbon, the carbon halogen bond or alkyl halide is polarised, the carbon atom bears a partial positive charge whereas the halogen atom bears a partial negative charge.

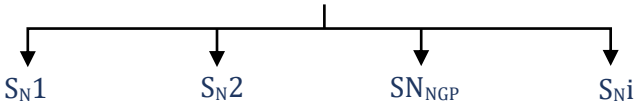


Replacement (displacement) of an atom or group by another atom or group in molecule is known as substitution reaction. If substitution reaction is brought about by a nucleophile, then it is known as nucleophilic substitution reaction. Generally, substitution takes place at sp^3 carbon.

$$L + Nu^{\ominus} \longrightarrow R—Nu + L^{\ominus}$$ (L – leaving group)

Nucleophilic substitution reactions can be classified as following:

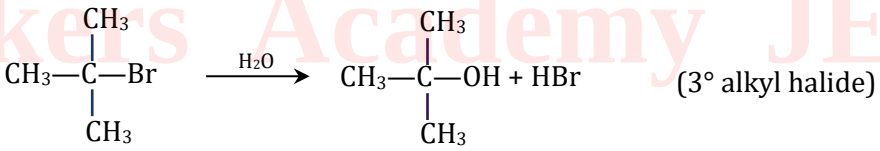
Nucleophilic substitutions reactions



Unimolecular nucleophilic substitution reaction (S_N1):

Mechanism : S_N1 in alkyl halides

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(3° alkyl halide)

Nucleophilic substitution involves two step process.

First step : Slow step involves ionisation to form carbocation. It is a rate determining step

$$R—L \longrightarrow R^{\oplus} + L^{\ominus}$$ (Slow)

Ionisation is always assisted by the solvent since energy necessary to break the bond is largely recovered by solvation of $R^{\oplus}$ and $L^{\ominus}$

Second step : Attack of nucleophile on carbocation to result into product, it is a fast step

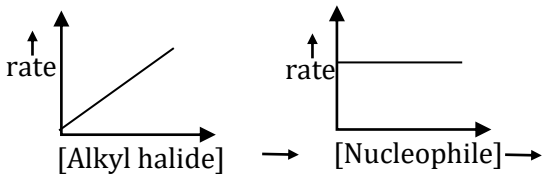
$$R^{\oplus} + Nu^{\ominus} \longrightarrow R—Nu$$ (fast)

1. **Characteristics of s_n1 reactions :**
   (a) It is unimolecular, two step process.
   (b) Carbocation intermediate is formed so rearrangement is possible in S_N1 reaction.

   $$\text{Rate} \propto [\text{Alkyl halide}]^1$$

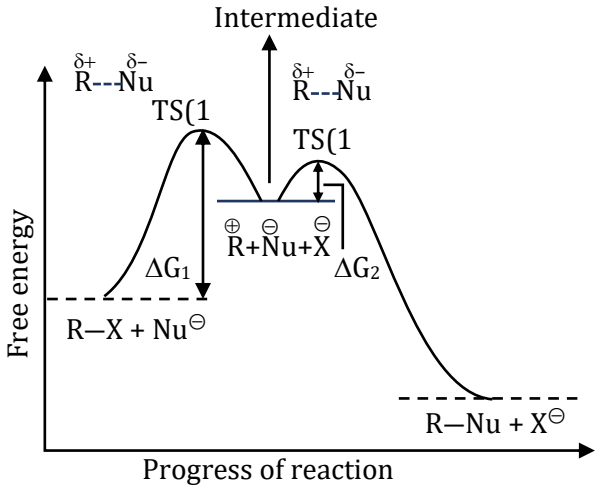
   $$\text{Rate} = K\,[\text{Alkyl halide}]^1$$

   Rate of S_N1 reaction is independent of concentration and reactivity of nucleophile, i.e., It is first order reaction



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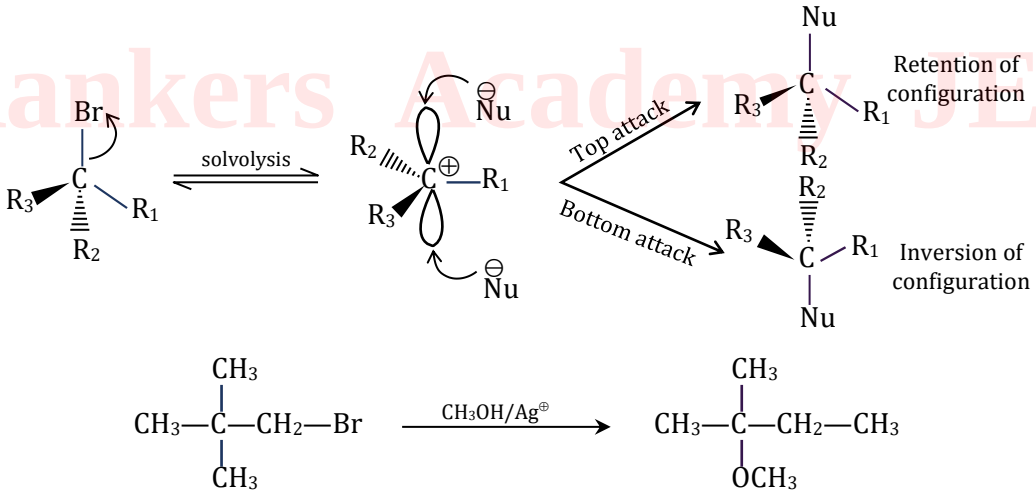
2. Energetics of the s_n1 :



3. Stereochemistry of s_n1 reactions :

In the S_N1 mechanism, the carbocation intermediate is sp^2 hybridized and planar, A nucleophile can attack on the carbocation from either face, if reactant is chiral than after attack of nucleophile, from both faces, both enantiomers are formed as the product, which is called racemisation.

4. Mechanism of racemisation (S_N1) :

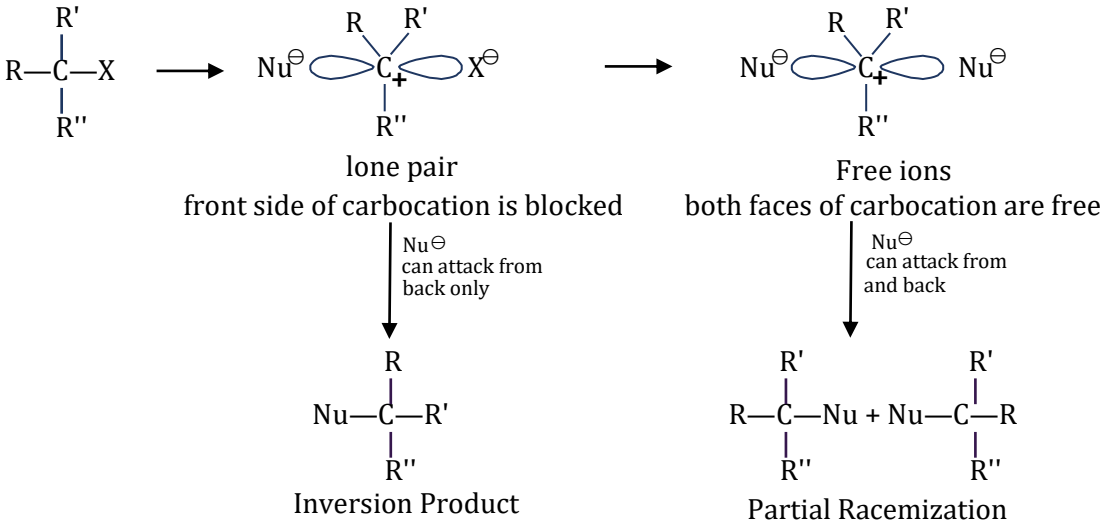


(Carbocation rearrangement)

If enantiomers are formed in equal amounts then reaction is said to proceed with complete racemisation. In the case of complete racemisation no optical activity is detected in the mixture. However, in practice the expected complete racemisation is rarely observed. The products usually consists of more inversion product than retention product.

In most of the case the product has usually of 5-20 % inverted product and 80-95% racemised species. Thus reaction proceed with partial racemisation and some inversion. This may be explained by considering that the attack by a nucleophile occurs before the leaving groups has completely departed from the neighbourhood of the carbocation, thus, to a certain extent the leaving group shields the front side of the carbocation from attack. Consequently, the backside attack is preferred to some extent resulting in the inversion of configuration.

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The more stable is the carbocation, the greater is the proportion of racemisation. This is because in such cases the leaving group gets time to leave neighbourhood of the carbocation before the attack by a nucleophile occurs, thus, there is almost equal facility for attack from either side of the carbocation plane leading to a greater degree of racemisation. In solvolysis reaction, more nucleophilic is solvent, greater is the proportion of inversion.

5. Factors affecting reactivity in S_N1 reactions:

(A) Effect of substrate structure : Since rds involves formation of carbocation hence all the factors which affect stability of carbocation, will also affect rate of S_N1 reactions.

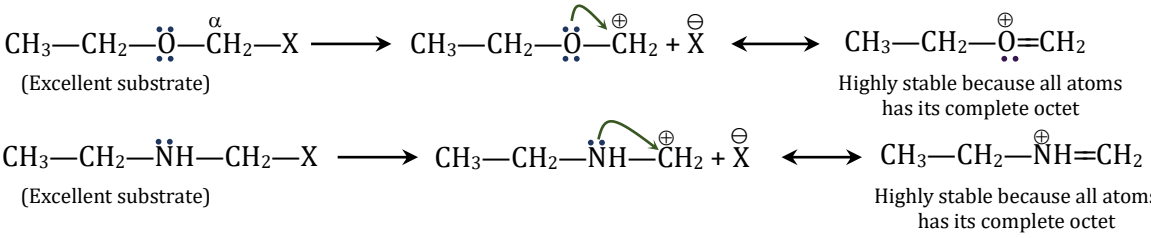
Decreasing order of reactivity of some substrates in S_N1 reactions

$Ar_3CX > Ar_2CHX > R_3CX > ArCH_2X > CH_2 = CH - CH_2 - X > R_2CHX > RCH_2X$

(a) Unsaturation at the α-carbon: In such cases S_N1 reactions are highly unfavoured because unsaturation at α carbon creates partial double bond character in C–X bond. That's why vinyl halides and aryl halides do not give S_N1 reactions.

(b) Unsaturation at the β-carbon: In such cases S_N1 reactions are highly favoured because formed carbocation is resonance stabilised

(c) Presence of heteroatom at α-carbon : If heteroatom is present at α position then it can stabilize formed carbocation by direct lone pair donation.



(d) Presence of carbonyl group on β-carbon: carbonyl group at β carbon destabilizes carbocation, that's why in such cases S_N1 is not possible.

(e) Substitution effect : Greater is the number of alkyl group at $C^{\oplus}$, more will be stability of formed arbocation. Hence rate of reaction will be faster

(B) Effect of Solvent : Greater is the ionising ability of the solvent more will be rate of S_N1.

Because to solvate cations and anions so effectively the use of a polar protic solvent will greatly increase the rate of ionisation of an alkyl halide in any S_N1 reaction. It does this because solvation stabilises the transition state leading to the intermediate carbocation and halide ion more than it does the reactant, thus the energy of activation is lower.

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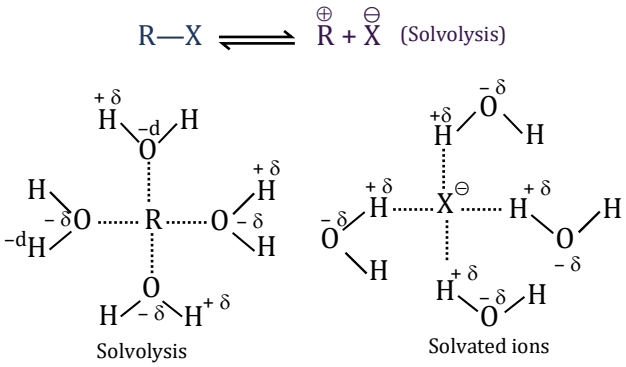


Table : Dielectric constants ($\in$) and ionisation rates of t-Butylchloride in common solvents.

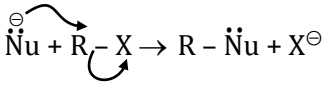
Solvent	$\in$	Relative rate
H_2O	80	8000
CH_3OH	33	1000
C_2H_5OH	24	200
$(CH_3)_2CO$	21	1
CH_3CO_2H	6	–

(C) Effect of leaving group : In the S_N1 reaction the leaving group begins to acquire a negative charge as the transition state is reached. Stabilisation of this developing negative charge at the leaving group stabilises the transition state and this lowers the free energy of activation and there by increases the rate of reaction. Leaving ability of halogen is $F^{\ominus} < Cl^{\ominus} < Br^{\ominus} < I^{\ominus}$
Hence greater is the leaving power of leaving group, more will be reactivity of substrate towards S_N1.

(D) Effect of the attacking nucleophile : RDS does not involve nucleophile or its nucleophilicity. Hence rate of S_N1 reactions are unaffected by the concentration and nature of the nucleophile. Hence neither nucleophile nor its nucleophilicity has any effect on rate of S_N1.
Weak, neutral, mostly solvents (protic) itself functions as nucleophiles in S_N1 reaction. So S_N1 reaction are termed as solvolysis reaction.

water $\rightarrow$ hydrolysis : $C_2H_5OH \rightarrow$ ethanolysis
$CH_3COOH \rightarrow$ acetolysis : $NH_3 \rightarrow$ ammonolysis

Some S_N reactions of alkyl halide $\rightarrow$



NOTE :

1. When base is weak but strongly nucleophilic toward carbon, then E2/S_N2 ratio is low but in the presence of strong base the E2/S_N2 ratio increases.
2. Rate of reaction increases as temperature increases. Thus the E_2/S_N2 ratio will increase with the increasing temperature.

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Illustration 10

Which among the following forms most stable carbocation:

(A) $(CH_3)_3C—Br$ (B) $(C_6H_5)_3C—Br$

(C) $(C_6H_5)_2CH—Br$ (D) $(C_6H_5)—CH_2—Br$

Ans. (B)

Solution:

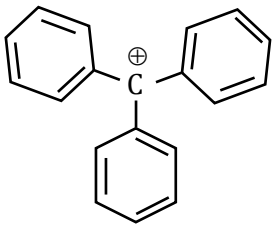
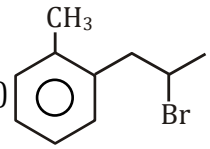
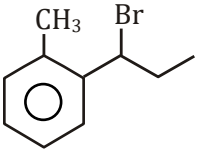
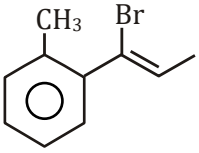
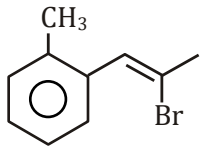
$(C_6H_5)_3\overset{\oplus}{C} \equiv$  most stable carbocation, due to maximum extent of resonance.

Illustration 11

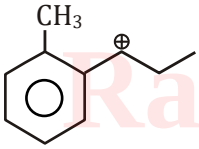
Which compound undergoes hydrolysis by the S_N1 mechanism at fastest rate?

(A)  (B)  (C)  (D) 

Ans. (B)

Solution:

Rate of hydrolysis by $S_N1 \propto$ stability of carbocation

 is most stable carbocation as it is stabilised by resonance.

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Illustration 12

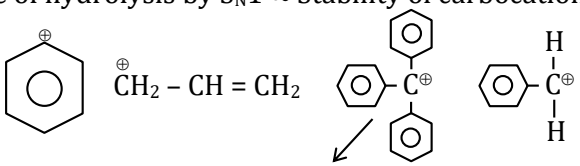
Which of the following compounds is most rapidly hydrolysed by S_N1 mechanism?

(A) C_6H_5Cl (B) $Cl–CH_2–CH=CH_2$ (C) $(C_6H_5)_3CCl$ (D) $C_6H_5CH_2Cl$

Ans. (C)

Solution:

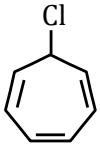
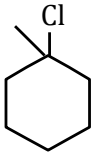
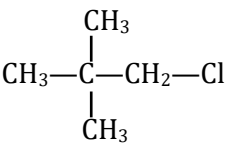
Rate of hydrolysis by $S_N1 \propto$ Stability of carbocation



Maximum extent of resonance

Illustration 13

Arrange the following compounds in decreasing order of reactivity towards S_N1 reaction :-

(a) $Ph–CH_2–Cl$ (b)  (c)  (d) 

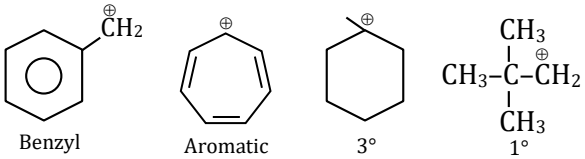
(A) a > c > b > a (B) c > d > b > a (C) a > b > c > d (D) b > a > c > d

Ans. (D)

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Solution:

Reactivity in $S_N1 \propto$ Stability of carbocation



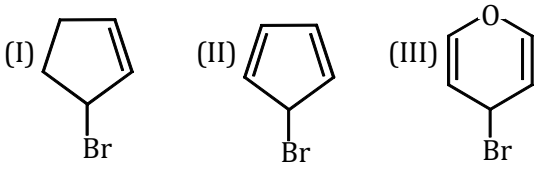
(a) (b) (c) (d)

Stability order : b > a > c > d

Reactivity order : b > a > c > d

Illustration 14

Among the bromides I–III given below, the order of reactivity by S_N1 reaction is:



(A) III > I > II (B) III > II > I (C) II > III > I (D) II > I > III

Ans. (A)

Solution:

Order of reactivity in $S_N1 \propto$ Stability of carbocation



(I) (II) (III)

Stability → aromatic > non-aromatic > anti-aromatic

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Bimolecular nucleophilic substitution reaction (S_N2):

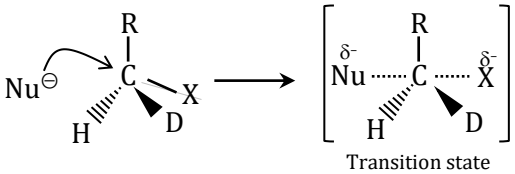
Nucleophilic substitution in which incoming group replaces leaving group in one step only.

Reaction does not involve formation of any intermediate.

S_N2 Reaction of Alkyl halide :

$$R - X + Nu^{\ominus} \rightarrow R - Nu + X^{\ominus}$$

Mechanism :



Transition state

1. **Characteristics of S_N2 reactions:**
   (A) Nucleophile attacks on the substrate from just opposite side to the leaving group.
   (B) Hybridisation of the carbon at which substitution occurs changes from sp^3 to sp^2 in the transition state.

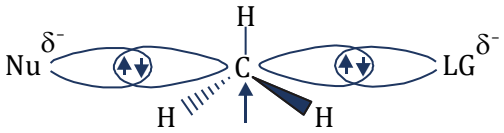
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(C) It is bimolecular, one step concerted process

rate ∝ [alkyl halide] [nucleophile]

rate = k [alkyl halide] [nucleophile]

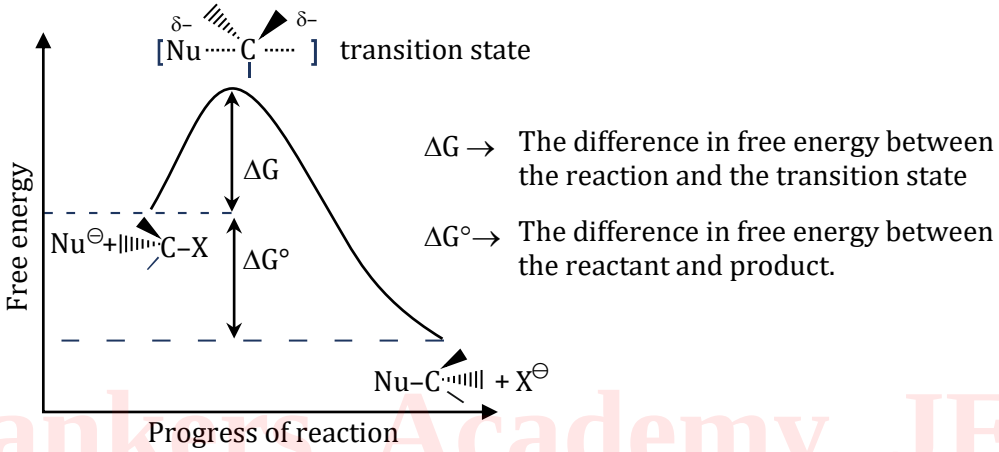
(D) Transition state has following structure



(Pentavalent carbon in sp^2 hybridised state)

Bond with nucleophile and leaving group are relatively long and weak in transition state.

2. Energetics of the reaction:



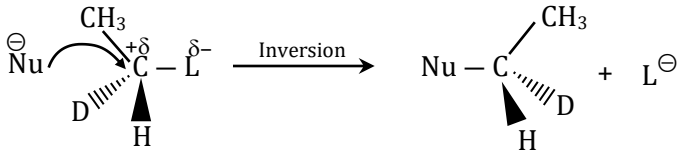
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Figure: A free energy diagrams for S_N2 reaction

No intermediates are formed in S_N2 reaction, the reaction proceeds through the formation of an unstable arrangement of atoms or group called transition state.

3. The stereochemistry of s_n2 reactions:

In S_N2 mechanism the nucleophile attacks from the back side, that is from the side directly opposite to the leaving group. This mode of attack causes an inversion of configuration at the carbon atom that is the target of nucleophilic attack. This inversion is also known as **Walden inversion**.



It is always not necessary that absolute configuration will change. Absolute configuration will invert when both incoming nucleophile and leaving group have same priority according to C.I.P rule

4. Factor's affecting the rate of S_N2 reaction:

Number of factors affecting the relative rate of S_N2 reaction, the most important factors are

(A) Effect of substrate structure:

(a) Alkyl groups at the α and β carbons: Since S_N2 reaction are very sensitive to steric hinderance, hence if alkyl groups are present at α and β carbon, then they will increase steric crowding that leads to decrement in rate of reaction.

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Reactivity order towards S_N2 reaction

$CH_3–X$ > 1° alkyl halide > 2° alkyl halide >> 3° alkyl halide (unreactive)

The important reason behind this order of reactivity is a steric effect. Very large and bulky groups can often hinder the formation of the required transition state and crowding raises the energy of the transition state and slows down reaction.

Relative rate of reactions of alkyl halide in S_N2 reaction.

Substituent	Compound	Relative rate
Methyl	CH_3X	30
1°	CH_3CH_2X	1
2°	$(CH_3)_2CHX$	0.02
Neopentyl	$(CH_3)_3CCH_2X$	0.00001
3°	$(CH_3)_3CX$	~ 0

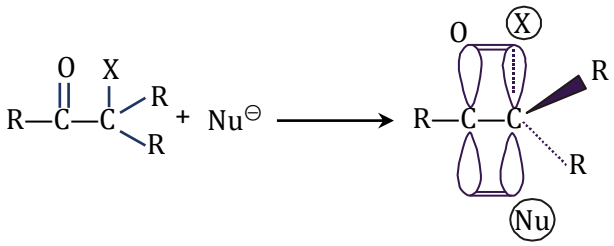
(b) Presence of unsaturation on β carbons : Presence of unsaturation on β carbon in primary alkyl halide increases rate of S_N2 , that's why allyl halide and benzyl halides are good substrate for S_N2

(c) Presence of hetero atom on α carbon : Lone pair present on hetero atom stabilizes transition state by delocalization. Hence rate of reaction increases

(d) Presence of carbonyl group at β carbon : Presence of carbonyl group at α–carbon makes a substrate favourable for S_N2 reactions it is due to p-orbitals on carbonyl group are parallel to the p-orbital in S_N2 transition state. In such cases reaction will be exceptionally faster A comparison of rate of relations towards S_N2

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Substrate	$CH_3–X$	$CH_2=CH–CH_2–X$	$C_6H_5–CH_2–X$	$CH_3–O–CH_2–X$	$C_6H_5–\overset{\overset{O}{\|}}{C}–CH_2X$
Relative ate	200	79	200	920	10^5



[Transition state]
Stabilisation of the transition state by p–orbital of the CO group

(B) Concentration and reactivity of the nucleophile:

(i) Since nucleophile is involved in expression of rate of S_N2. Hence on increasing concentration of nucleophile, rate of S_N2 increases.

(ii) As nucleophilicity of nucleophile increases rate of S_N2 reaction increases.

(iii) Anionic nucleophiles mostly give S_N2 reaction

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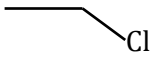
Some common nucleophiles listed in decreasing order of nucleophilicity in hydroxylic solvent	
Strong nucleophiles:	Moderate nucleophile
$(Ph)_3P$	$\overset{\ominus}{Br}$
$\overset{\ominus}{S}H$	NH_3
$I^{\ominus}$	$(CH_3)_2S$
$(CH_3-CH_2)_2NH$	$\overset{\ominus}{Cl}$
$-\overset{\ominus}{C}N$	$AcO^{\ominus}$
	Weak nucleophile :
$NH_2^{\ominus}$	$F^{\ominus}$
$HO^{\ominus}$	H_2O
$CH_3O^{\ominus}$	CH_3OH

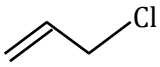
(C) The effect of the solvent:

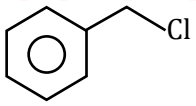
Polar aprotic solvent have crowded positive centre, so they do not solvate the anion appreciably therefore the rate of S_N2 reactions increased when they are carried out in polar aprotic solvent.

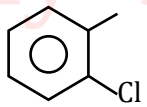
Illustration 15

In which of the following replacement of $Cl^{\ominus}$ is most difficult ?

(A) 

(B) 

(C) 

(D) 

Ans. (D)

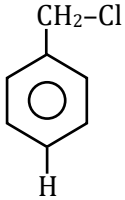
Solution:

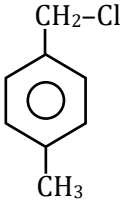
Cl attached to Benzene ring develops partial double bond character due to resonance.

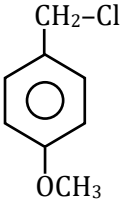
So, bond breaking is difficult for substitution. Also, π-electron cloud of benzene ring repels the nucleophile, so S_N2 becomes more difficult

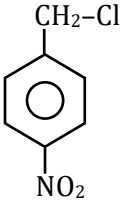
Illustration 16

Which of the following is most reactive toward S_N2.

(A) 

(B) 

(C) 

(D) 

Ans. (D)

Solution:

In S_N2 reaction, the electron density at electrophilic centre increases

So, electron withdrawing group will stabilise transition state.

More stable the transition state, faster will be the reaction

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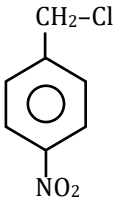
So  molecule will be more reactive towards S_N2 reactions as it contains electron withdrawing – NO_2 group.

Illustration 17

Which of the following is the correct order of decreasing S_N2 reactivity?

(A) $(CH_3)_2CHCl > (CH_3)_3CCl > CH_3CH_2Cl$

(B) $CH_3CH_2Cl > (CH_3)_3CCl > (CH_3)_2CHCl$

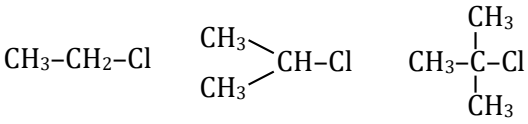
(C) $CH_3CH_2Cl > (CH_3)_2CHCl > (CH_3)_3CCl$

(D) $(CH_3)_3CCl > (CH_3)_2CHCl > CH_3CH_2Cl$

Ans. (C)

Solution:

Order of S_N2 reactivity : 1° > 2° > 3°



So, correct order: $CH_3CH_2Cl > (CH_3)_2CHCl > (CH_3)_3CCl$

Illustration 18

When the concentration of alkyl halide is tripled and the concentration of $OH^{\ominus}$ ion is reduced to half, the rate of S_N2 reaction increases by X times. Report your answer as 10 X.

Ans. (15)

Solution:

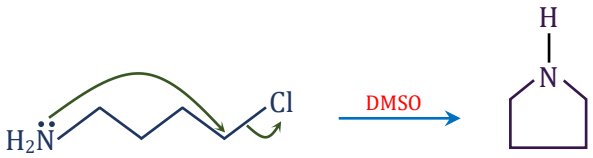
Rate of $S_N2 \propto [R - X]\,[Nu^{\ominus}]$

$$\frac{r_2}{r_1} = \frac{[3RX]\left[\frac{1}{2}OH^{\ominus}\right]}{[RX]\;[OH^{\ominus}]}$$

$r_2 = 1.5\ r_1$

S_NNGP Nucleophilic substitution by Neighbouring Group Participation:

When substrate contains a group with an unshared pair of electron beta to the leaving group or sometimes further away then it is called neighboring group participation or S_NNGP.

Example : 

Characteristics of reaction:

S_NNGP is two step reaction-

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Step I : Intramolecular S_N2 which is NGP step
Step II : Intermolecular S_N2
Compound having lone pair at 3rd, 5th or 6th atom with respect to leaving group show the NGP step (Intramolecular S_N2).
Nucleophile must be anti to leaving group to show NGP step within the same molecule.
S_NNGP is faster than S_N1 & S_N2

Reaction of alcohol with phosphorus halides:

$$R-OH+PCl_5 \longrightarrow R-Cl+POCl_3+HCl$$

$$3R-OH+PCl_3 \longrightarrow 3R-Cl+H_3PO_3$$

$$3R-OH+PBr_3 \longrightarrow 3R-Br+H_3PO_3$$

$$3R-OH+PI_3 \longrightarrow 3R-I+H_3PO_3$$

Phosphorous halides are prepared by treating red phosphorous and halogen. The advantage of using phosphorous halides is that the reaction does not involve carbocation intermediate so, it is free from rearrangement.
PBr_3 and PI_3 are less stable, thus for bromides and Iodides, (P + Br_2) Or (P + I_2) mixture is used.

Mechanism for Reaction with phosphorus trihalides

$$3R{-}OH + PX_3 \xrightarrow{(PX_3=PCl_3,\ PBr,\ PI_3)} 3R{-}X + H_3PO_3$$

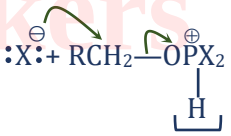
Step:1



Protonated alkyl dihalophosphite

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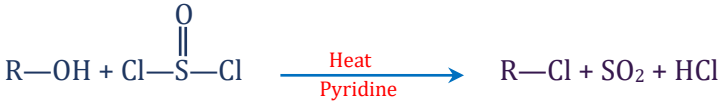
Step:2



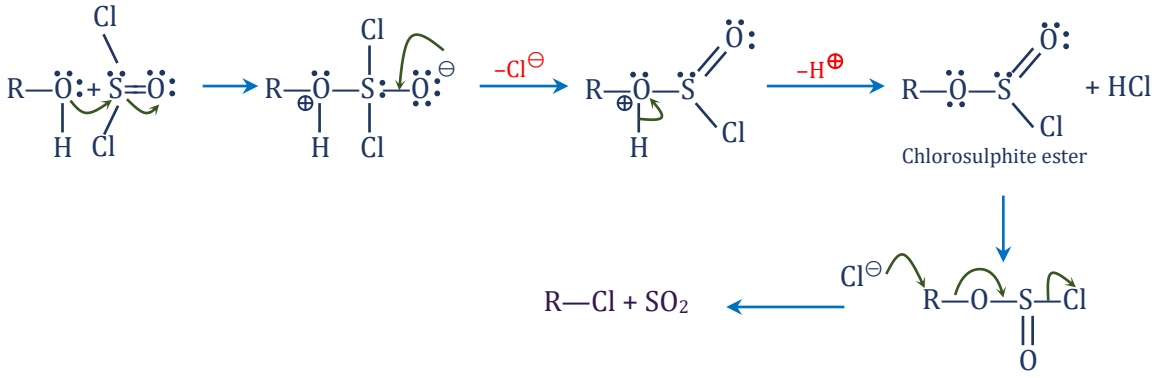
A good leaving group $\longrightarrow$ $RCH_2X + HOPX_2$

Reaction with thionyl chloride in presence of pyridine:

Thionyl chloride ($SOCl_2$) is often the best reagent for converting an alcohol to an alkyl chloride. The by products (gaseous SO_2 and HCl) leave the reaction mixture and ensure that there can be no reverse reaction.



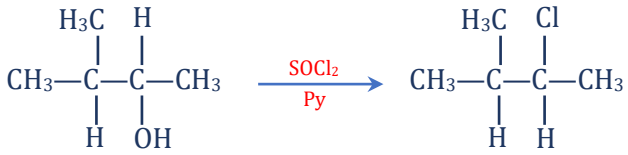
Mechanism :



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In the first step, the nonbonding electrons of the hydroxy oxygen atom attack the electrophilic sulphur atom of thionyl chloride. A chloride ion is expelled a proton and gives test of chloro sulphite ester.
Second step is an S_N2 mechanism

Example :

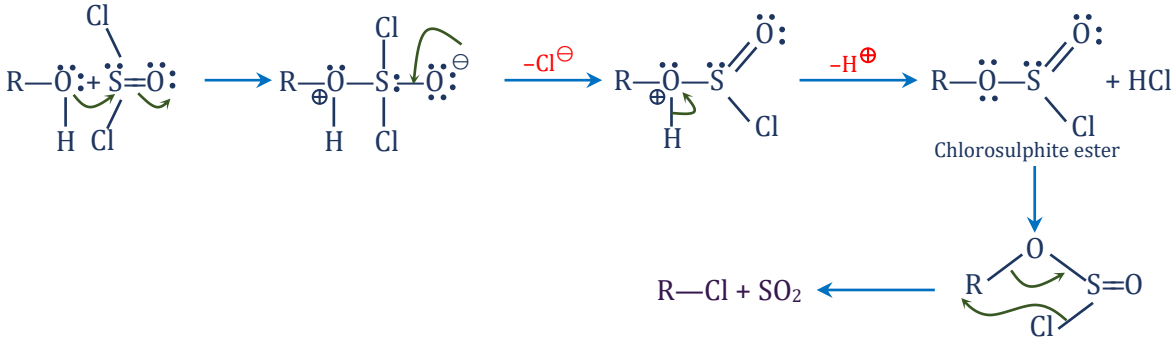


Reaction with thionyl chloride:

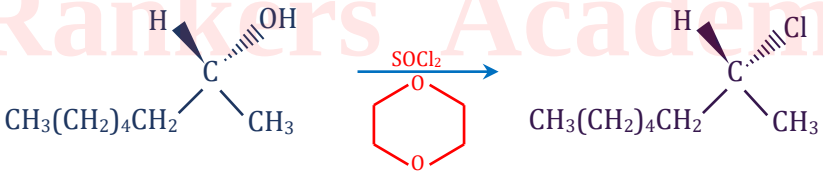
$$ROH + SOCl_2 \longrightarrow RCl + SO_2 + HCl$$

In this mechanism an internal nucleophile attacks from the same side of leaving group, means retention of configuration. It is an S_Ni mechanism, where i means internal.

Mechanism :



Example :

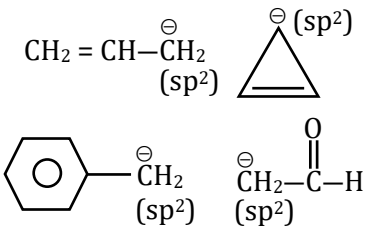


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Characteristics of carbanion :

(i) Negatively charged species
(ii) $8e^-$ in valence shell
(iii) Complete octet
(iv) Electron rich species
(v) Act as nucleophile
(vi) Lewis base (because it can donate l.p)
(vii) 3σ-bond, one lone pair, zero vacant orbital
(viii) zero unpaired electron, Diamagnetic in nature
(ix) Hybridisation sp^3, Trigonal pyramidal

Note : When –ve charge is in resonance then sp^2.

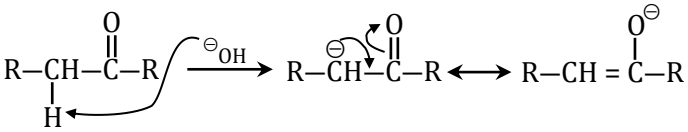


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(x) No rearrangement
(xi) stability of carbanion is depends on resonance effect inductive effect.

Some general method of generation of carbanion

(a) By attack of base on $\underset{H}{\overset{|}{C}}-x=y$ system (E.N. y > x)

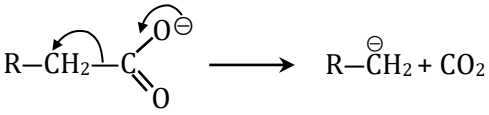


(b) By attack of base on ≡C—H :

(i) $CH\equiv C-H \xrightarrow{\ominus NH_2} CH\equiv C^{\ominus}$

(ii) $CH_3-C\equiv C-H \xrightarrow{\ominus NH_2} CH_3-C\equiv C^{\ominus}$

(c) By decarboxylation of carboxylate ion



(d) In the form of organometallic compound

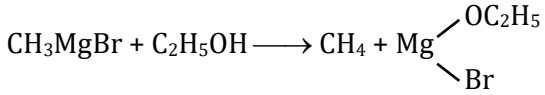
(i) $CH_3-Cl \xrightarrow{Mg/ether} \overset{\delta-}{C}H_3\overset{\delta+}{M}gCl$

(ii) $2CH_3-CH_2-I \xrightarrow[CuI]{Li} (CH_3-CH_2)_2CuLi$ (Gilman Reagent)

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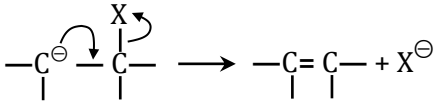
Some general reactions of carbanion

(a) Attachment of $H^{\oplus}$



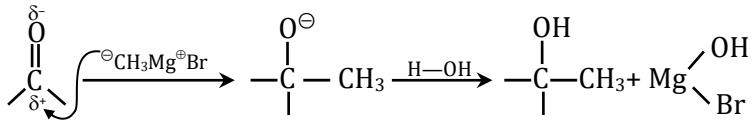
(b) Loss of Leaving Group

In the system $\overset{\ominus}{C}-\overset{|}{\underset{|}{C}}-X$ carbanion losses leaving group to form a π bond

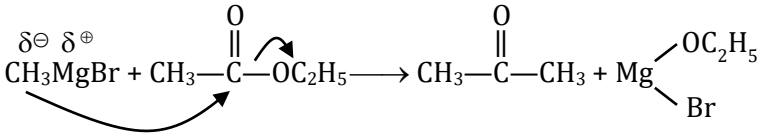


(c) Addition Reaction

Carbanion gives Nucleophilic addition reaction :

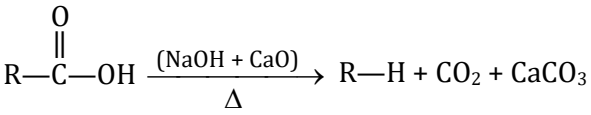


(d) Nucleophilic substitution via addition elimination.



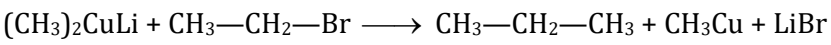
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(e) Soda lime decarboxylation



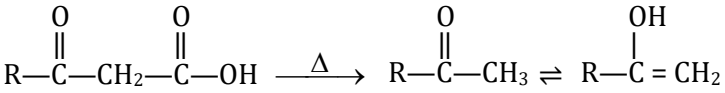
$$R-\overset{\overset{O}{\|}}{C}-OH \xrightarrow[\Delta]{(NaOH + CaO)} R-H + CO_2 + CaCO_3$$

(f) Corey house synthesis



$$(CH_3)_2CuLi + CH_3-CH_2-Br \longrightarrow CH_3-CH_2-CH_3 + CH_3Cu + LiBr$$

(g) Heating effect



$$R-\overset{\overset{O}{\|}}{C}-CH_2-\overset{\overset{O}{\|}}{C}-OH \xrightarrow{\Delta} R-\overset{\overset{O}{\|}}{C}-CH_3 \rightleftharpoons R-\overset{\overset{OH}{|}}{C}=CH_2$$

Illustration 19

Most stable carbanion is :

(A) $\overset{\ominus}{C}H_2-CH_2-NO_2$

(B) $\overset{\ominus}{C}H_2-CH_2-CN$

(C) $\overset{\ominus}{C}H_2-CH_2-COOH$

(D) $\overset{\ominus}{C}H_2-CH_2-CH_2-CH_2-F$

Ans. (A)

Solution:

(A) $\overset{\ominus}{C}H_2-CH_2-NO_2$

–I group

(B) $\overset{\ominus}{C}H_2-CH_2-CN$

–I group

(C) $\overset{\ominus}{C}H_2-CH_2-COOH$

–I group

(D) $\overset{\ominus}{C}H_2-CH_2-CH_2-CH_2-F$

–I group

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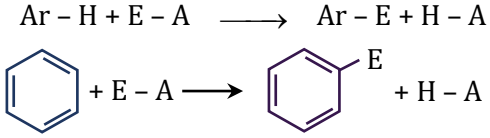
–I effect is highest for $-NO_2$, so most stable carbanion is $\overset{\ominus}{C}H_2-CH_2-NO_2$

Electrophilic Aromatic Substitution Reaction:

Introduction:

Aromatic hydrocarbons are known generally as arenes. An aryl group is one derived from an arene by removal of a hydrogen atom and its symbol is Ar—. Thus, arenes are designated ArH just as alkanes are designated RH.

The most characteristic reactions of benzenoid arenes are the substitution reactions that occur when they react with electrophilic reagents. These reactions are of the general type shown below.

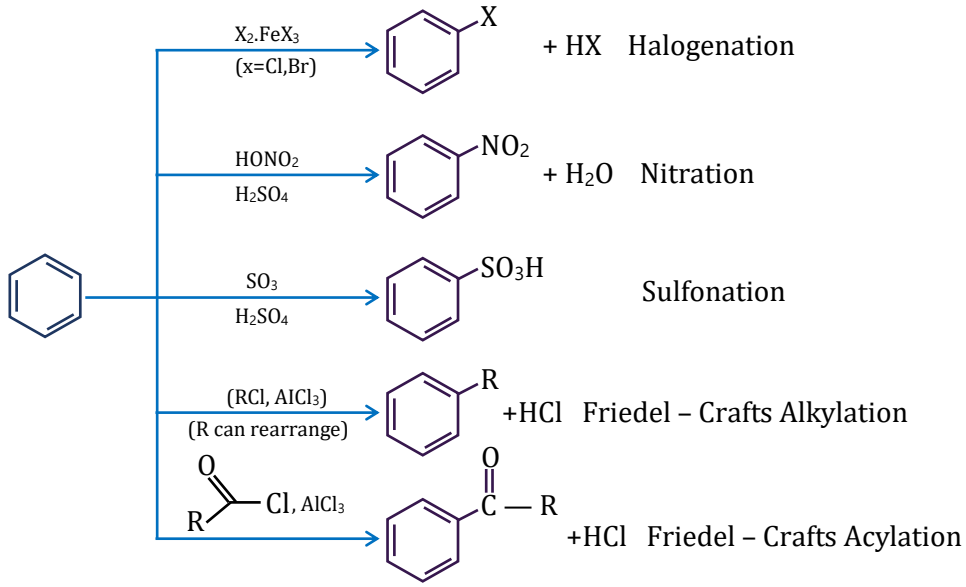


$$Ar-H + E-A \longrightarrow Ar-E + H-A$$

The electrophiles are either a positive ion ($E^{\oplus}$) or some other electron- deficient species with a large partial positive charge. For example, benzene can be brominated when it reacts with bromine in the presence of $FeBr_3$. Bromine and $FeBr_3$ react to produce positive bromine ions, $Br^{\oplus}$. These positive bromine ions act as electrophiles and attack the benzene ring, replacing one of the hydrogen atoms in a reaction that is called an electrophilic aromatic substitution (EAS).

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Electrophilic aromatic substitutions allow the direct introduction of a wide variety of groups into an aromatic ring, and because of this they provide synthetic routes to many important compounds. The five electrophilic aromatic substitutions that we shall study in this chapter are outlined in figure.



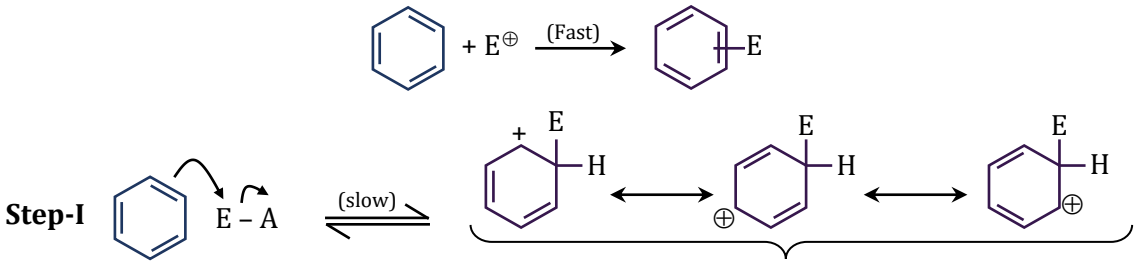
Electrophilic Aromatic Substitution Reactions

(1) A General Mechanism for electrophilic Aromatic substitution: Benzene is susceptible to electrophilic attack primarily because of its exposed π-electrons. In this respect benzene resembles an alkene, for in the reaction of an alkene with an electrophilic the site of attack is the exposed π bond.

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We saw in however, that benzene differs from an alkene in a very significant way. Benzene's closed shell of six π electrons gives it a special stability. So, although benzene is susceptible to electrophilic attack, it undergoes substitution reactions rather than addition reactions. Substitution reactions allow the aromatic sextet of π electrons to be regenerated after attack by the electrophile was occurred. We can see how this happens of we examine a general mechanism for electrophilic aromatic substitution.

Once the electrophilic, $E^{\oplus}$ is generated in the reaction, it enters into some kind of a weak interaction with the π cloud of benzene ring leading to the formation of a π - complex. This π - complex is a donor-acceptor type of a complex, benzene being the donor and electrophile, the acceptor. These adducts are known as charge transfer complexes. In the complex that benzene forms with bromine, it has been shown that the halogen molecule is located centrally and at right angles to the plane of the benzene ring.

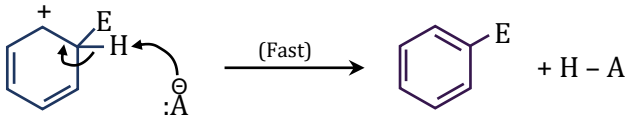


In step 1 the electrophile takes two electrons of the six- electron π system to form a σ bond to one carbon atom of the benzene ring. Formation of this bond interrupts the cyclic system of π - electrons, because in the formation of the arenium ion the carbon that forms a bond to the

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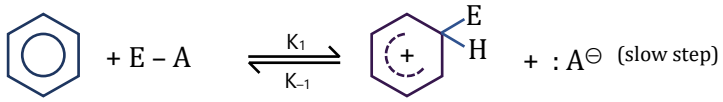
electrophile becomes sp^3 hybridized and therefore, no longer has an available p-orbital. Now only five carbon atoms of the ring are still sp^2 hybridized and still have p-orbitals. A calculated electrostatic potential map for the arenium ion formed by electrophilic addition of bromine to benzene indicates that positive charge is distributed in the arenium ion ring (figure) just as was shown in the contributing resonance structures.

In step 2 a proton is removed from the carbon atom of the arenium ion that bears the electrophile. The two electrons that bonded this proton to carbon becomes a part of the π system. The carbon atom that bears the electrophile becomes sp^2 hybridized again and a benzene derivative with six fully delocalized π electrons is formed. We can represent step 2 with any one of the resonance structures for the arenium ion.

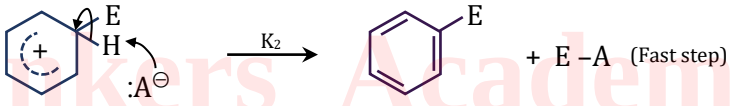
Step-II 

(The proton is removed by any of the bases present, for example, by the anion derived from the electrophile).

Kekule structure are more appropriate for writing mechanisms such as electrophilic aromatic substitution because they permit the use of resonance theory, it can be described however, using the modern formula for benzene in the following way.

Step-I 

Arenium ion

Step-II 

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Where k_1 and k_{-1} is the rate constant of the forward and backward reactions in step-I, k_2 is the rate constant of the step-II.

Isotope Effects:

The next question which comes up is how do we ascertain the presence of two discrete steps in the mechanism and also that the formation of σ - complex is the rate determining step. Answer to this question can be obtained from the study of kinetic isotope effect. If the rate of a reaction depends on a step which involves breaking of a C—H bond, then a kinetic isotope effect (K_H/K_D) of 6 to 7 is expected. Absence of any significant isotope effect in aromatic electrophilic substitution (except sulphonation) suggests that the proton is lost in the fast step, subsequent to rds. We see that the isotope effect study has provided two pieces of important information regarding the mechanism. Firstly, it has shown that the reaction takes place in two steps and secondly, that the first step is slower than the second step.

(2) Nitration:

Nitration reaction is generally carried out with a mixture of concentrated nitric acid and sulphuric acid.

The reagents which cause nitration are called nitrating agents.

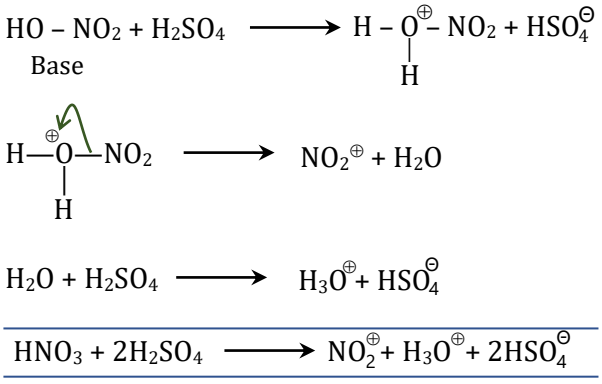
The electrophile involved in nitration reaction is nitronium ion $\left(NO_2^{\oplus}\right)$.

Mechanism:

Generation of electrophile from nitrating agent.

(a) In a mixture of nitric acid and sulphuric acid, an acid base reaction takes place in which nitric acid acts as the base.

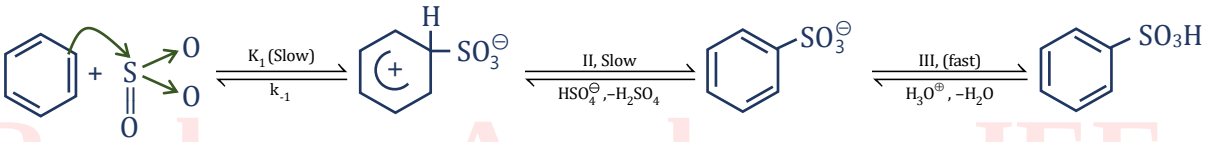
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(3) Sulphonation:

Sulphonation is another synthetically important reaction. It is often accomplished with concentrated sulphuric acid or fuming sulphuric acid containing excess of SO_3 or chlorosulphonic acid, HSO_3Cl.

It is believed that the electrophile varies with the reagent, though in all cases SO_3, is involved either free or along with a carrier, like in H_2SO_4 ($SO_3 + H_3O^{\oplus}$) or $H_2S_2O_7$. Sulphur trioxide is generated from sulphuric acid as follows $2H_2SO_4 \rightleftharpoons HSO_4^{\ominus} + H_3O^{\oplus} + SO_3$. The mechanism of sulphonation of benzene is given below:



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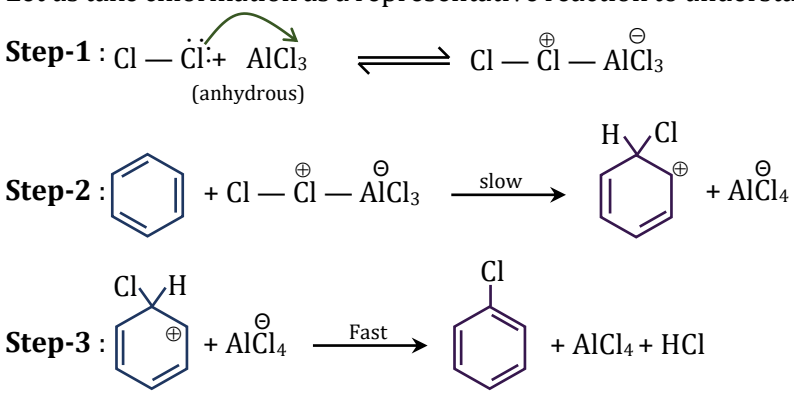
Sulphonation is different from other aromatic electrophilic substitution reactions, Firstly, it is reversible and secondly, it shows some amount of isotope effect which is totally absent in other cases.

(4) Halogenation :

Halogenation of an aromatic ring is a synthetically important reaction. It takes place in the presence of varied reaction condition depending on the reactivity of the aromatic ring. For very reactive aromatic compounds in polar solvents, the molecular halogens themselves may acts as electrophiles. In the case of nonpolar solvents, halogenation is catalysed by a Lewis acid like $AlCl_3$, or $FeCl_3$, Reactivity of halogens has the following order,

$$I_2 < Br_2 < Cl_2$$

Let us take chlorination as a representative reaction to understand the mechanism of halogenation.



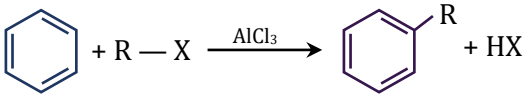
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Bromination follows a similar mechanism. As said above, Iodine is weakest of the three halogens and even in the presence of a Lewis acid, it can halogenate reactive aromatics only. Therefore, in most other cases iodine-monochloride is used in the presence of Lewis acid, $ZnCl_2$.

(5) Fridel-Crafts Alkylation:

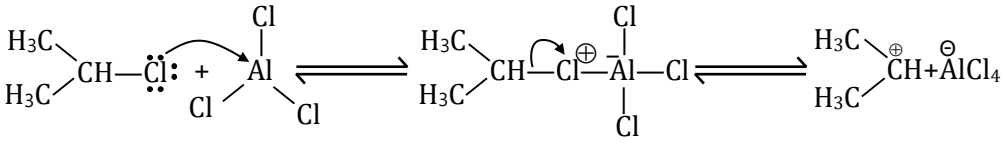
Alkyl halides react with benzene in presence of aluminium chloride to yield alkyl benzenes. Alkylation of benzene with alkyl halides in the presence of aluminium chloride was discovered by Charles Friedel and James. M. Crafts in 1877.

A general equation for a friedel-Crafts alkylation reaction is the following

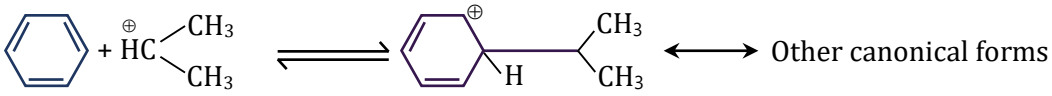


Mechanism for the Reaction :

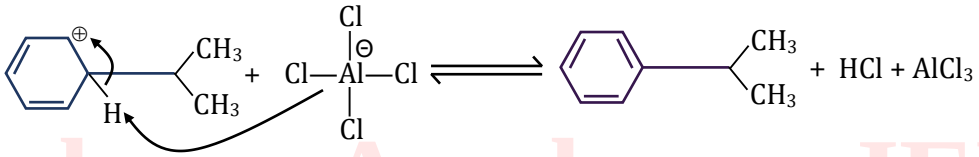
Step-I



Step-II



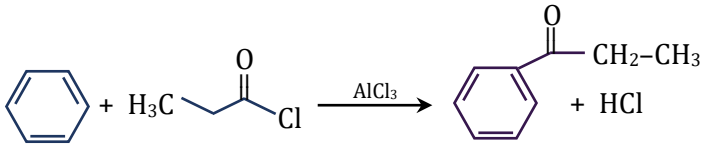
Step-III



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(6) Friedel-crafts Acylation:

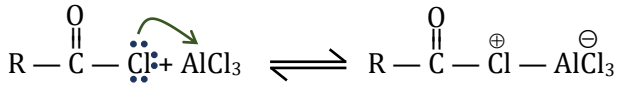
The Friedel-Crafts acylation reaction is an effective means of introducing an acyl group into an aromatic ring. The reaction is often carried out by treating the aromatic compound with an acyl halide.



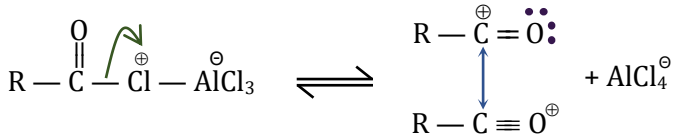
Mechanism of the Reaction:

In most Friedel-Crafts acylation reactions the electrophile appears to be an acylium ion formed from an acyl halide in the following way.

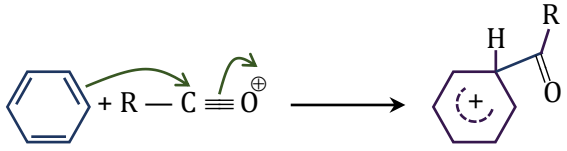
Step-I



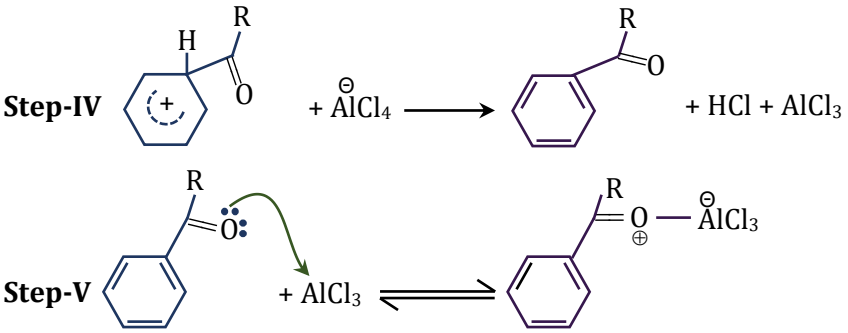
Step-II



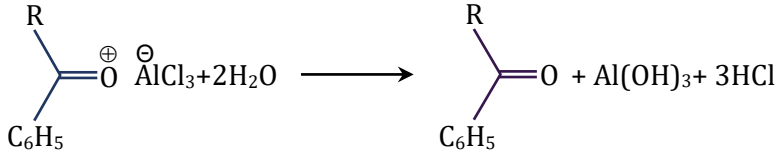
Step-III



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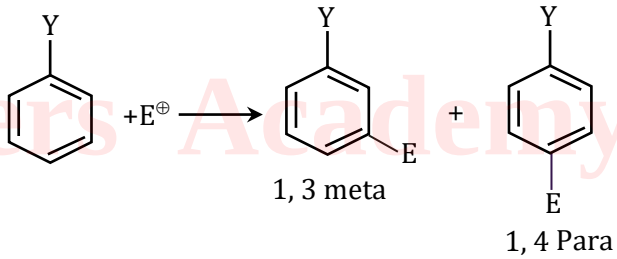


In the last step, $AlCl_3$ (a Lewis acid) forms a complex with the ketone (a Lewis base). After the reaction is over, treating the complex with water liberates the ketone.



(7) Orientation and reactivity in electrophilic aromatic substitution :

In the case of benzene we get just one monosubstituted product in electrophlic aromatic substitution as all the positions are equivalent. However, for substitution in a compound which already has a group attached to the ring, three disubstituted products, (1, 2), (1, 3) (1, 4) commonly called ortho, meta and para meaning next to, between and opposite respectively are possible. These are abbreviated as o, m and P.



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Reactivity:

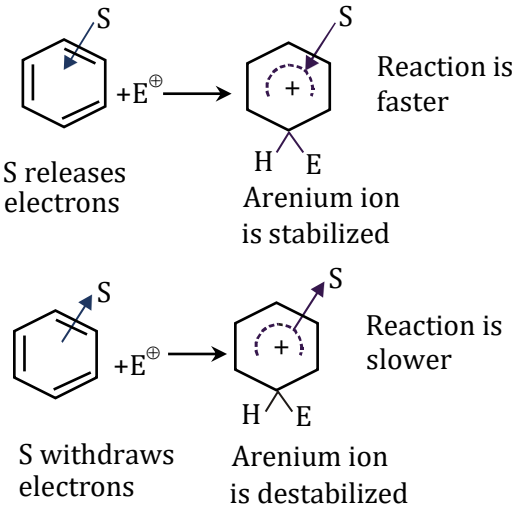
We have now seen that certain groups activate the benzene ring toward electrophilic substitutions, while other groups deactivate the ring. We mean that an aromatic compound with an activating group reacts faster in electrophilic substitutions than benzene. When we say that a group deactivates the ring, we mean that an aromatic compound with a deactivating group reacts slower than benzene.

We have also seen that we can account for relative reaction rates by examining the transition state for the rate-determining steps. Any factor that increases the energy of the transition state relative to that of the reactants decreases the relative rate of the reaction. It does this because it increases the free energy of activation of the reaction. In the same way, any factor that decreases the energy of the transition state relative to that of activation and increases the relative rate of the reaction.

The rate-determining step in electrophilic substitutions of substituted benzene compounds is the step that results in the formation of arenium ion. We can write the formula for a substituted benzene in a generalized way if we use the letter S to represent any ring substituent including hydrogen, (if S is hydrogen, the compound is benzene). We can also write the structure for the arenium ion in the way shown here. By this formula we mean that S can be in any position -ortho, meta or para - relative to the electrophile, E. Using these conventions, we can write the rate-determining step for electrophilic aromatic substitution in the following way.

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When we examine this step for a large number of reactions, we find that the relative rates of the reactions depend on whether S withdraws or release electrons. If S is an electron-releasing group (relative to hydrogen), the reaction occurs faster than the corresponding reaction of benzene. If S is an electron withdrawing group, the reaction occurs slower than that of beznene.



It appears, that the substituent (S) must affect stability of the transition state relative to that of the reactants. Electron-releasing groups apparently make the transition state more stable, while electron withdrawing groups make it less stable. that is entirely reasonable, because the transition state resembles the arenium ion, and the arenium ion is a delocalized carbocation.

The effects of substituents on the reactivity of a benzene ring toward electrophilic substitution:

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Strong activating groups : —NH_2, —NHR, —NR_2, —OH, —OR

Moderately activating groups : —NHCR, — OCR

Weakly activating groups :—R, —Ar, —$CH{=}CR_2$

Weakly deactivating groups :—F, —Cl, —Br, —I

Ortho/para directing

Moderately deactivating groups : —CHO, —COR, —COOR, —COOH, —COCl

Strong deactivating groups : —C≡N, —SO_3H, — $\overset{\oplus}{N}H_3$, — $\overset{\oplus}{N}H_2R$, — $\overset{\oplus}{N}HR_2$, — $\overset{\oplus}{N}R_3$, —NO_2

Meta directing

(8) Heterocyclic aromatic compound :

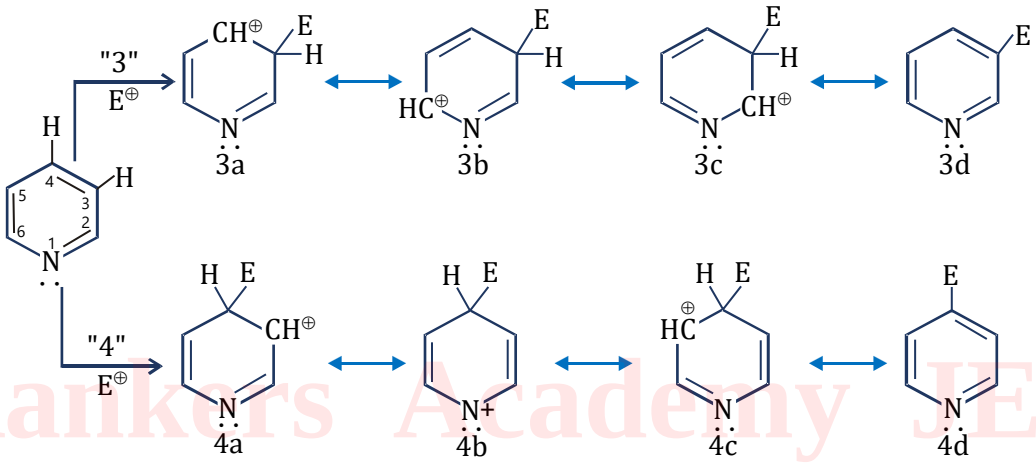
This is a class of compounds with aromatic properties, in which one or more of the ring carbons has been replaced by a heteroatom (an atom other than carbon) such as nitrogen, oxygen or sulphur. Some of these compounds preserve the six-membered ring and are therefore called benzenoid aromatic. In other, the (4n +2) π-electron reside in a five- membered ring, so these are called non-benzenoid. A key difference between these two classes of aromatic compounds is that in non-benzenoid heteroaromatic compounds a formally non-bonding pair of electrons on the heteroatom becomes part of the (4n +2) π-electron of the aromatic π-system, whereas in benzenoid aromatic compounds, the non-bonding pairs on heteroatoms are normally separated from the aromatic system. Here are a few examples :

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Benzenoid heteroaromatic compounds: Pyridine, Quinoline, Pyrimidine

Benzenoid heteroaromatic compounds: Pyrrole, Furan, Thiophene

Electrophilic Substitution in Pyridine:
In pyridine the attack may takes place at "3" or "4" position in general.



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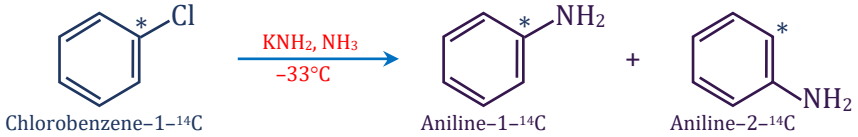
At first glance the two sets of resonance structures look quite similar. But notice how in 4b (one of the resonance structures of the intermediate that results from attack at the 4th position) the valence of nitrogen is left unfilled, an unfavourable situation not encountered in the intermediate formed by attack at the 3rd position. Again, it is the electronegativity of the nitrogen that reduces the reactivity of pyridine (relative to benzene) toward electrophiles.

Illustration 20
How many species are ortho para directing
(i) –OH (ii) –COOH (iii) $-CONH_2$
(iv) $-CH_3$ (v) $-O^{\ominus}$ (vi) $-\overset{\oplus}{N}H_3$

Ans. (3)
Solution:
ortho para directing species are –OH, $-CH_3$, $-O^{\ominus}$

The Elimination-Addition Mechanism of Nucleophilic Aromatic Substitution (Benzyne):

(i) Very strong base such as sodium or potassium amide react with aryl halide, even those without electron withdrawing substituents to give products corresponding to nucleophilic substitution of halide by the base.

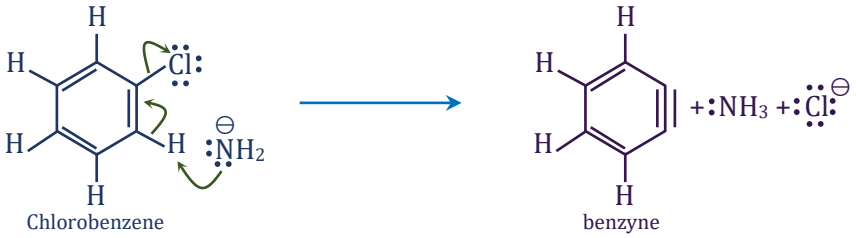


Chlorobenzene–1–^{14}C Aniline–1–^{14}C Aniline–2–^{14}C

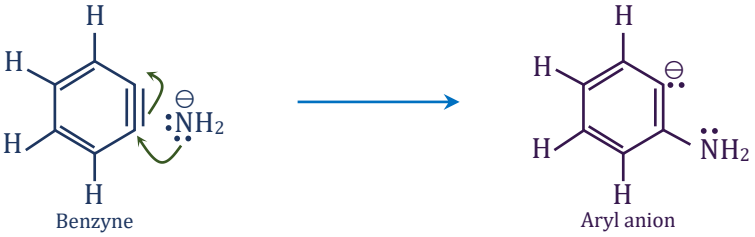
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Mechanism:

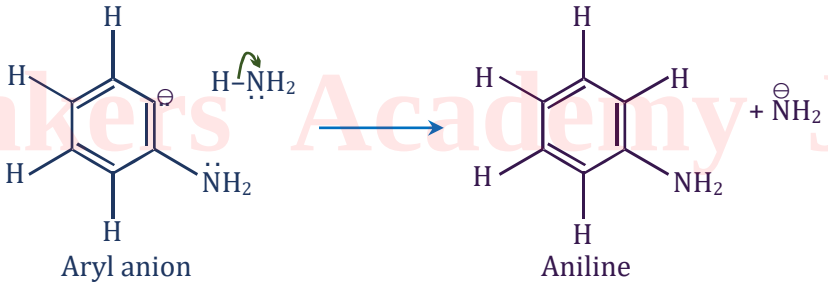
Step-1: Elimination stage; Amide ion is a very strong base and brings about the dehydrohalogenation of chlorobenzene by abstracting a proton from the carbon adjacent to the one that bears the leaving group. The product of this step is an unstable intermediate called benzyne.



Step-2 : Beginning of addition phase ; Amide ion acts as a nucleophile and adds to one of the carbons of the triple bond. The product of this step is a carbanion.



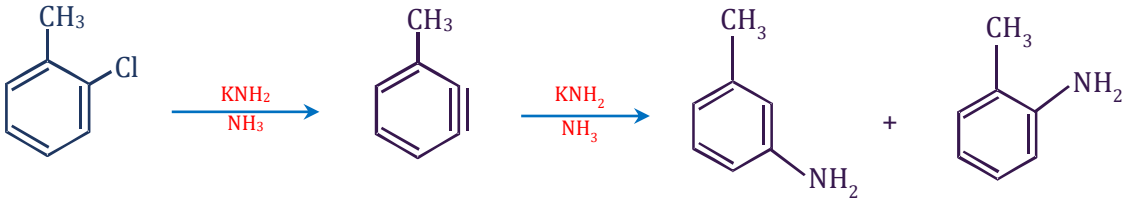
Step-3 : Completion of addition phase ; The aryl anion abstracts a proton from the ammonia used as the solvent in the reaction.



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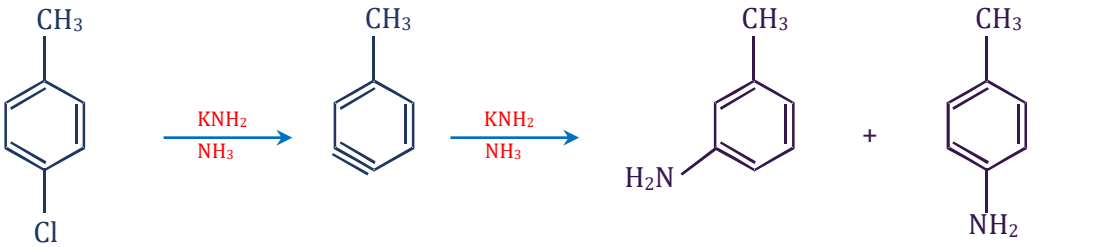
NOTE : The sp^2 orbital in the plane of the ring in benzyne are not properly aligned for good overlap thus π bonding is weaker than alkyne.

(i)



O-chloro toluene — 3-Methylbenzyne — m-Methylaniline — o-Methylaniline

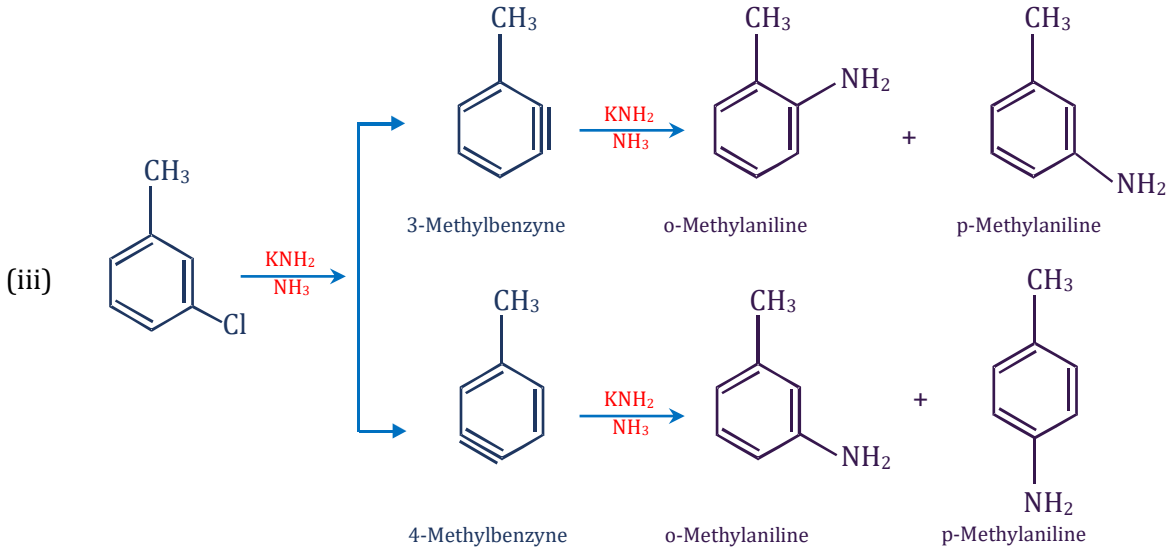
(ii)



p-Chlorotoluene — 4-methylbenzyne — m-Methylaniline — p-Methylaniline

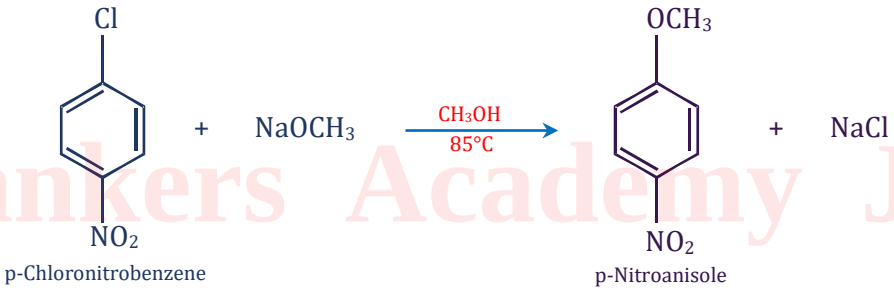
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(iii)



Nucleophilic Aromatic Substitution by the addition Elimination Mechanism:

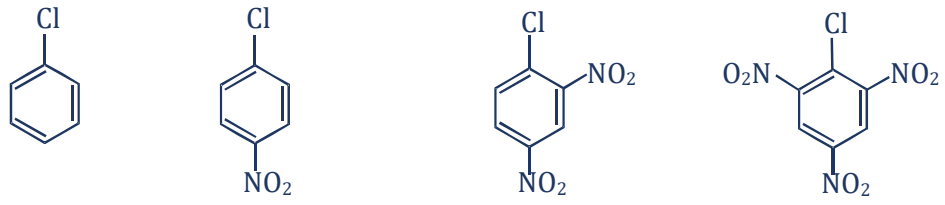
(i) The generally accepted mechanism for nucleophilic aromatic substitution in nitro-substituted aryl halides.



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(ii) An ortho - nitro group exert a comparable rate–enhancing effect, m-chloronitrobenzyne while much more reactive than chlorobenzyne itself, is thousand of times less reactive than either o-or p-chloronitrobenzene.

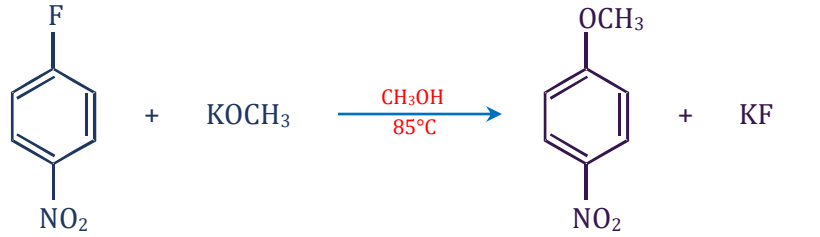
(iii) The effect of o- & p-nitro substituents is cumulative, as the rate data for substitution with methoxide ion in a series of nitro-substituted chlorobenzene derivative demonstrate increasing rate of reaction as :



	Chlorobenzene	1–Chloro–4–nitrobenzene	1–Chloro–2, 4–dinitrobenzene	2–Chloro–1, 3, 5–trinitrobenzene
Relative rate :	1.0	7×10^{10}	2.4×10^{15}	(too fast to measure)

(iv) In contrast to nucleophilic substitution in alkyl halides, where alkyl fluorides are exceedingly unreactive, aryl fluorides undergo nucleophilic substitution readily when the ring bears an o-or a p-nitro group.

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p-Fluoro nitrobenzene | Potassium methoxide | p-Nitroanisole (93%) | Potassium fluoride

(v) Indeed, the order of leaving group reactivity in nucleophilic aromatic substitution is the opposite of that seen in aliphatic substitution.

(vi) Fluoride is the best reactive leaving group in nucleophilic aromatic substitution, iodide the least reactive.



Relative reactivity towards sodium methoxide in methanol (50°C)

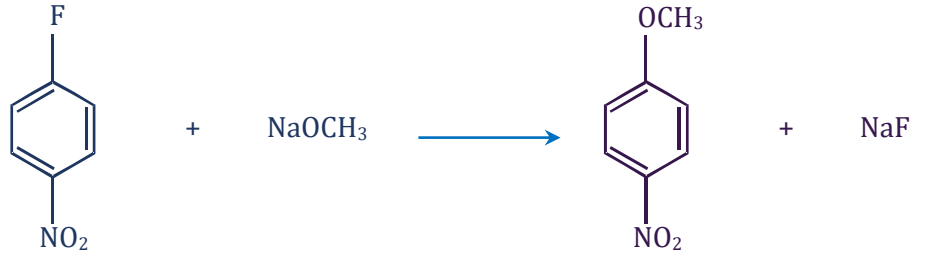
X =	F	Cl	Br	I
	312	1.0	0.8	0.4

(vii) Kinetic studies of many of the reactions described in the section have demonstrated that they follow a second-order rate law.

Rate = k[aryl halide] [nucleophile]

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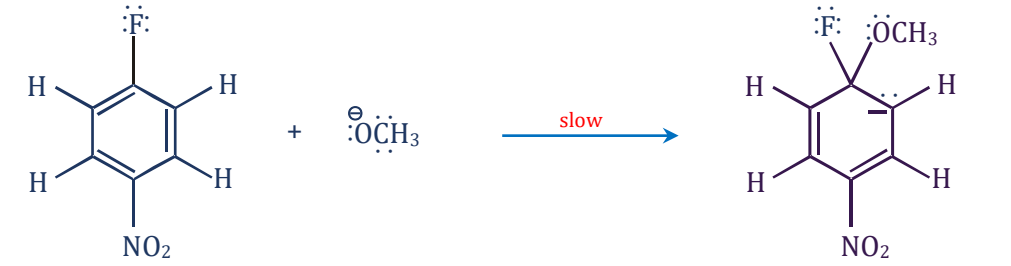
(viii) Second order kinetics is usually interpreted in terms of a bimolecular rate determining step.



p-Fluoronitrobenzene | Sodium methoxide | p-Nitroanisole | Sodium Fluoride

Mechanism :

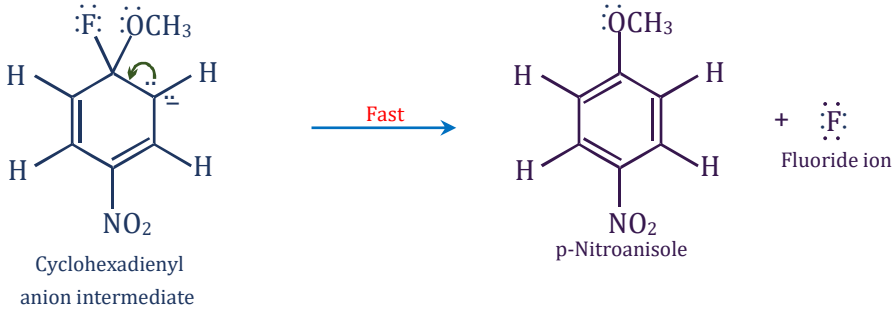
Step-1 Addition stage. The nucleophile, in this case methoxide ion, adds to the carbon atom that bears the leaving group to give a cyclohexadienyl anion intermediate.



p-Fluoronitrobenzene | Methoxide ion | Cyclohexadienyl anion intermediate

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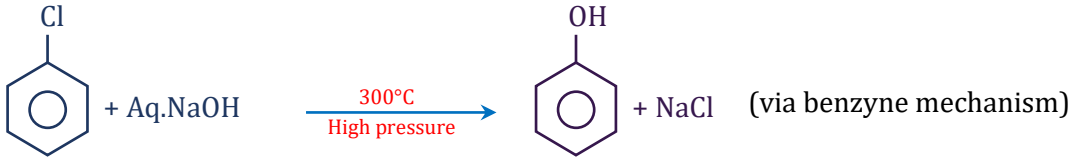
Step-2 Elimination stage. Loss of halide form the cyclohexadienyl intermediate restores the aromaticity of the ring and gives the product of nucleophilic aromatic substitution.



Cyclohexadienyl anion intermediate

p-Nitroanisole

(a) Dow process :



(via benzyne mechanism)

Presence of deactivating group in ortho and para position makes the nucleophilic substitution easier.

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